

LAPORAN TUGAS BESAR

IF2123 Aljabar Linier dan Geometri



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I. Pendahuluan

A. Latar Belakang

Aljabar linier merupakan salah satu dasar penting dalam bidang komputasi dan digunakan dalam berbagai permasalahan, seperti simulasi rekayasa, pemodelan aliran energi, analisis data, dan pemodelan sistem. Berbagai permasalahan tersebut dapat melibatkan operasi matriks dan sistem persamaan linear dengan ukuran serta tingkat kompleksitas yang cukup besar sehingga diperlukan proses perhitungan yang terstruktur.

Pada Tugas Besar 1 mata kuliah IF2123 Aljabar Linear dan Geometri, dikembangkan sebuah pustaka berbasis bahasa pemrograman Java yang dirancang untuk mengimplementasikan berbagai metode perhitungan matriks. Pustaka yang dikembangkan mencakup penyelesaian Sistem Persamaan Linier (SPL), perhitungan determinan, pencarian invers matriks, interpolasi polinomial, Natural Cubic Spline Interpolation, dan regresi spline kubik.

Selain implementasi metode perhitungan, pustaka juga perlu dirancang untuk menghasilkan skenario pengujian. Program utama menggunakan Command Line Interface (CLI) dan mendukung masukan manual maupun file *.txt*. Hasil perhitungan pula dapat ditampilkan pada layar dan dapat disimpan ke dalam file *.txt*.

Melalui tugas besar ini, implementasi algoritma aljabar linier dapat diterapkan pada pengembangan perangkat lunak yang modular dan terstruktur. Dengan demikian, tugas besar ini menjadi sarana untuk menerapkan konsep-konsep aljabar linier ke dalam persoalan dunia nyata.

B. Tujuan

Tujuan tugas besar ini sebagai berikut:

1. Mengimplementasikan berbagai metode aljabar linier.
Metode aljabar linier meliputi penyelesaian Sistem Persamaan Linier (SPL), perhitungan determinan, pencarian matriks balikan, interpolasi polinomial, Natural Cubic Spline Interpolation, dan Regresi Spline Kubik menggunakan bahasa pemrograman Java.
2. Mengembangkan pustaka (*library*) yang modular dan terstruktur.
3. Membangun program utama berbasis CLI dan dapat menerima masukan melalui *keyboard* maupun file *.txt*, serta menampilkan dan menyimpan hasil perhitungan.

II. Dasar Teori

A. Sistem Persamaan Linier (SPL)

Sistem Persamaan Linier (SPL) merupakan kumpulan berhingga dari persamaan-persamaan linier dengan variabel yang sama. Sebuah SPL dengan m persamaan dan n variabel memiliki bentuk umum [1] :

$$\begin{array}{rcl}
a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & = & b_1 \\
a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & = & b_2 \\
\vdots & & \vdots \\
a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & = & b_m
\end{array}$$

(Gambar: bentuk umum sistem persamaan linear)

atau dalam bentuk $Ax = b$, maupun dalam bentuk matriks augmented seperti berikut :

$$[A \mid b] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right]$$

(Gambar: matriks augmented)

Terdapat tiga kemungkinan solusi dari SPL, yaitu solusi tunggal, banyak solusi, dan tidak ada solusi. Secara umum, metode yang dapat digunakan untuk memperoleh solusi tersebut antara lain eliminasi Gauss, Gauss-Jordan, matriks balikan, dan kaidah cramer.

B. Determinan Matriks

Matriks persegi memiliki determinan yang merupakan nilai skalar (angka tunggal) yang menunjukkan faktor skala perubahan luas atau volume dari transformasi geometri yang dilakukan oleh matriks [1]. Determinan dari matriks A dinotasikan sebagai $\det(A)$ atau $|A|$. Perhitungan matriks melalui,

a. Metode Reduksi Baris / OBE

Pada metode ini, matriks A dilalui proses Operasi Baris Elementer (OBE) untuk hingga menjadi matriks segitiga (bawah atau atas)

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \rightarrow \begin{bmatrix} a'_{11} & a'_{12} & \dots & a'_{1n} \\ 0 & a'_{22} & \dots & a'_{2n} \\ \vdots & \vdots & \dots & a'_{3n} \\ 0 & 0 & 0 & a'_{nn} \end{bmatrix}$$

(Gambar: matriks persegi $\rightarrow$ OBE $\rightarrow$ matriks segitiga bawah)

Rumus determinan A,

$$\det(A) = (-1)^p a'_{11} a'_{22} \dots a'_{nn} / (k_1 k_2 \dots k_n), \text{ dengan}$$

p = banyaknya operasi pertukaran baris di dalam OBE

$k_1 k_2 \dots k_n =$ perkalian baris-baris matriks di dalam OBE

b. Metode Ekspansi Kofaktor

Matriks persegi berukuran $n \times n$ memiliki minor entri a_{ij} (M_{ij}) atau determinan upa-matriks (submatrix) yang elemen-elemennya tidak berada pada baris i dan kolom j . Kofaktor (C_{ij}) adalah nilai minor suatu elemen matriks persegi yang telah dikalikan dengan tanda positif atau negatif berdasarkan posisi baris dan kolomnya.

Kofaktor entri a_{ij} ,

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Dengan menggunakan kofaktor, determinan matriks A dapat dihitung dengan salah satu persamaan berikut:

1. Secara baris

$$|A| = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}$$

$$|A| = a_{21}C_{21} + a_{22}C_{22} + \dots + a_{2n}C_{2n}$$

$\vdots$

$$|A| = a_{n1}C_{n1} + a_{n2}C_{n2} + \dots + a_{nn}C_{nn}$$

2. Secara kolom

$$|A| = a_{11}C_{11} + a_{21}C_{21} + \dots + a_{n1}C_{n1}$$

$$|A| = a_{12}C_{12} + a_{22}C_{22} + \dots + a_{n2}C_{n2}$$

$\vdots$

$$|A| = a_{1n}C_{1n} + a_{2n}C_{2n} + \dots + a_{nn}C_{nn}$$

C. Matriks Balikan (Invers)

Matriks balikan atau invers dari suatu matriks persegi A adalah matriks A^{-1} yang memenuhi hubungan:

$$[A \mid I] \rightarrow [I \mid A^{-1}] \quad [1]$$

dengan I merupakan matriks identitas. Suatu matriks memiliki invers jika dan hanya jika determinannya tidak sama dengan nol, yaitu $\det(A) \neq 0$ [1]. Matriks yang memiliki invers disebut matriks invertible atau nonsingular.

a. Metode Augmentasi Gauss-Jordan

Pada metode ini, matriks A digabungkan dengan matriks identitas I sehingga membentuk matriks augmented:

$$[A \mid I]$$

Selanjutnya dilakukan operasi baris elementer dan eliminasi Gauss-Jordan hingga bagian kiri berubah menjadi matriks identitas:

$$[A \mid I] \rightarrow [I \mid A^{-1}]$$

Matriks yang berada di sebelah kanan setelah proses tersebut selesai merupakan matriks balikan A^{-1} . Apabila bagian kiri tidak dapat diubah menjadi matriks identitas karena terdapat pivot bernilai nol tanpa baris pengganti, maka matriks tersebut tidak memiliki invers.

b. Metode Adjoint

Metode adjoint menggunakan hubungan antara determinan, matriks kofaktor, dan matriks balikan. Matriks kofaktor C diperoleh dengan menghitung kofaktor untuk setiap elemen matriks A . Matriks adjoin merupakan transpose dari matriks kofaktor:

$$\text{adj}(A) = C^T$$

Jika $\det(A) \neq 0$, matriks balikan dapat dihitung dengan:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) \quad [2]$$

D. Interpolasi Polinomial

Interpolasi polinomial merupakan metode untuk menentukan suatu polinomial yang melalui sejumlah titik data yang diberikan. Berdasarkan $n + 1$ titik data:

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \quad [2]$$

Koefisien $a_0, a_1, \dots, a_n$ ditentukan sehingga polinomial melewati seluruh titik data, sehingga berlaku:

$$P_n(x_i) = y_i$$

untuk setiap titik data yang diberikan. Derajat polinomial ditentukan berdasarkan jumlah titik data dikurangi satu. Untuk memperoleh koefisien polinomial, setiap titik data disubstitusikan ke bentuk umum polinomial. Dari proses tersebut diperoleh Sistem Persamaan Linear (SPL) [1]:

$$\begin{aligned}
a_0 + a_1x_0 + a_2x_0^2 + \dots + a_nx_0^n &= y_0 \\
a_0 + a_1x_1 + a_2x_1^2 + \dots + a_nx_1^n &= y_1 \\
&\vdots \\
a_0 + a_1x_n + a_2x_n^2 + \dots + a_nx_n^n &= y_n
\end{aligned}$$

SPL tersebut kemudian diselesaikan menggunakan metode eliminasi Gauss untuk memperoleh nilai setiap koefisien polinomial. Setelah koefisien diperoleh, terbentuk persamaan interpolasi $P(x)$ yang dapat digunakan untuk memperkirakan nilai y pada suatu nilai x di dalam domain interpolasi.

E. Natural Cubic Spline Interpolation

Interpolasi Spline Kubik adalah metode interpolasi mengkonstruksi kurva mulus dari sekumpulan titik-titik diskrit [1]. Perbedaan dengan polinomial adalah kelengkungan kurva yang tetap stabil dan kontinu hingga turunan kedua.

Untuk kumpulan titik, $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, dibagi menjadi $n - 1$ segmen, dengan syarat titik data terdefinisi dan terurut. Segmen merupakan potongan polinomial kubik pada suatu subinterval.

Diberikan $n + 1$ titik data $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ dengan $x_0 < x_1 < \dots < x_n$. Titik-titik tersebut membentuk n subinterval $[x_i, x_{i+1}]$. Pada setiap subinterval didefinisikan satu polinom kubik

$$\begin{aligned}
S_i(x) &= a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3, \quad [3] \\
&\text{batasan } x_i \leq x \leq x_{i+1}, \\
&i = 0, 1, \dots, n - 1
\end{aligned}$$

Setiap segmen memiliki 4 koefisien (a_i, b_i, c_i, d_i) , sehingga terdapat $4n$ koefisien yang harus ditentukan.

$$\begin{aligned}
\text{Turunan pertama: } S_i'(x) &= b_i + 2c_i(x - x_i) + 3d_i(x - x_i)^2 \\
\text{Turunan kedua: } S_i''(x) &= 2c_i + 6d_i(x - x_i)
\end{aligned}$$

Pada knot-nya, $(x - x_i)$ bernilai 0, sehingga $S_i''(x)$ adalah $2c_i$.

Syarat Spline Kubik Natural [1]:

- $S_i(x)$ merupakan polinom kubik pada $[x_i, x_{i+1}]$, untuk $i = 0, 1, \dots, n - 1$.
- $S_i(x_i) = y_i$ dan $S_i(x_{i+1}) = y_{i+1}$, untuk $i = 0, 1, \dots, n - 1$.
- $S_{i+1}(x_{i+1}) = S_i(x_{i+1})$, untuk $i = 0, 1, \dots, n - 2$.
- $S_{i+1}'(x_{i+1}) = S_i'(x_{i+1})$, untuk $i = 0, 1, \dots, n - 2$.
- $S_{i+1}''(x_{i+1}) = S_i''(x_{i+1})$, untuk $i = 0, 1, \dots, n - 2$.
- $S''(x_0) = S''(x_n) = 0$ (syarat natural).

Persamaan $S(x)$ melalui langkah-langkah penyelesaian:

1. Menentukan koefisien a_i

Dari syarat b:

$$a_i = y_i, i = 0, 1, \dots, n$$

2. Menghitung lebar setiap subinterval

$$h_i = x_{i+1} - x_i, i = 0, 1, \dots, n - 1$$

3. Menentukan koefisien c_i melalui sistem persamaan linear

Substitusi syarat b, d, dan e ke bentuk $S_i(x)$ menghasilkan persamaan yang hanya memuat c_i :

$$h_{i-1}c_{i-1} + 2(h_{i-1} + h_i)c_i + h_ic_{i+1} = (3/h_i)(a_{i+1} - a_i) - (3/h_{i-1})(a_i - a_{i-1}), \\ i = 1, 2, \dots, n - 1$$

Syarat natural (syarat f) memberikan: $c_0 = 0$ $c_n = 0$

Persamaan-persamaan tersebut membentuk SPL berukuran $(n + 1) \times (n + 1)$ dengan matriks koefisien tridiagonal. Matrix diselesaikan dengan eliminasi Gauss-Jordan untuk memperoleh $c_0, c_1, \dots, c_n$.

4. Menentukan koefisien b_i dan d_i

Dari syarat e:

$$d_i = (c_{i+1} - c_i) / (3h_i), i = 0, 1, \dots, n - 1$$

Dari syarat b:

$$b_i = (a_{i+1} - a_i)/h_i - h_i(2c_i + c_{i+1})/3, i = 0, 1, \dots, n - 1$$

5. Menyusun persamaan setiap segmen

Koefisien a_i, b_i, c_i, d_i disubstitusikan ke:

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3, x \in [x_i, x_{i+1}]$$

6. Menghitung nilai turunan kedua pada setiap knot

Pada knot x_i berlaku $(x - x_i) = 0$, sehingga:

$$S''(x_i) = 2c_i, i = 0, 1, \dots, n$$

Sesuai syarat natural, $S''(x_0) = S''(x_n) = 0$.

7. Mengevaluasi nilai interpolasi $S(x)$

Untuk x dalam domain interpolasi $x_0 \leq x \leq x_n$, tentukan indeks segmen i yang memenuhi $x_i \leq x \leq x_{i+1}$, kemudian hitung:

$$S(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

F. Regresi Spline Kubik

Regresi spline kubik merupakan pemodelan hubungan non-linear antara dua variabel yang memiliki potongan-potongan polinomial berderajat tiga. Potongan-potongan tersebut disambung pada titik-titik yang disebut dengan knot (simpul), $\epsilon_1, \epsilon_2, \dots, \epsilon_K$.

Ruang fungsi spline kubik dengan K simpul internal $\epsilon_1, \epsilon_2, \dots, \epsilon_K$ memiliki ruang vektor berdimensi $M = 4 + K$, di mana 4 merupakan jumlah koefisien polinomial kubik standar. Ruang vektor tersebut memiliki fungsi basis, yaitu polinomial global dan fungsi terpancung per simpul. Fungsi polinomial global yang dimiliki adalah $1, X, X^2, X^3$. Sementara itu, fungsi

terpancung dapat ditemukan dengan $(X - \epsilon_k)^3$, untuk $k = 1, \dots, K$. Menggunakan $(u)_+ = \max(0, u)$, didapatkan pemodelannya, yaitu :

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \sum_k \beta_{3+k} (x - \epsilon_k)_+^3$$

Untuk N sampel data titik $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$ dengan kondisi jumlah data melebihi jumlah basis. Setiap titik memberikan satu persamaan dalam bentuk matriks:

$$y = X\beta + \epsilon$$

dimana y merupakan vektor observasi, β merupakan vektor parameter bobot, dan X merupakan matriks desain.

$X\beta$ merupakan proyeksi ortogonal y ke ruang kolom X , sehingga vektor residu $\epsilon = y - X\beta$ tegak lurus terhadap setiap kolom X :

$$X^T(y - X\beta) = 0$$

Sehingga, mampu diperoleh persamaan normal:

$$(X^T X)\beta = X^T y$$

III. Implementasi

A. Struktur Kelas

a. Kelas Matrix

Kelas Matrix merupakan struktur data dasar yang digunakan untuk menyimpan dan memanipulasi matriks. Matriks direpresentasikan menggunakan array dua dimensi bertipe double.

Atribut utama:

1. rows : menyimpan jumlah baris matriks.
2. cols : menyimpan jumlah kolom matriks.
3. data : menyimpan elemen-elemen matriks dalam bentuk `double[][]`.

Metode:

1. `Matrix(int rows, int cols)` : membuat matriks dengan ukuran tertentu.
2. `inputMatrix()` : menerima masukan matriks dari pengguna.
3. `getRows()` dan `getCols()` : memperoleh dimensi matriks.
4. `getValue(row, col)` : memperoleh nilai pada posisi tertentu.
5. `set(row, col, value)` : mengubah nilai elemen matriks.
6. `partialPivoting(...)` : melakukan pemilihan dan pertukaran baris berdasarkan elemen pivot.
7. `multiplyRow(...)` : mengalikan suatu baris dengan konstanta.
8. `addRowbyRow(...)` : melakukan operasi penjumlahan suatu baris dengan kelipatan baris lain.

9. `printMatrix()` : menampilkan matriks.

b. Kelas SPL

Kelas SPL digunakan untuk menemukan solusi dari sistem persamaan linear menggunakan metode eliminasi Gauss, Gauss-Jordan, matriks balikan, dan kaidah cramer.

Metode:

1. `methods(int inputMethod, Matrix m)` : Memilih metode eliminasi untuk memperoleh solusi SPL.
2. `eliminasiGauss(Matrix data)` : Mencari solusi SPL menggunakan metode eliminasi Gauss.
3. `eliminasiGaussJordan(Matrix data)` : Mencari solusi SPL menggunakan metode eliminasi Gauss
4. `eliminasiInverse(Matrix data)` : Mencari solusi SPL menggunakan metode matriks balikan.
5. `kaidahCramer(Matrix data)` : Mencari solusi SPL menggunakan metode kaidah cramer.
6. `printSPL(Matrix data)` : mencetak SPL.

c. Kelas ModuleDeterminan

Kelas ModuleDeterminan digunakan untuk melakukan perhitungan determinan matriks, terutama menggunakan metode ekspansi kofaktor baris atau kolom dan reduksi baris

Metode:

1. `ekspansiKofaktorBaris(int x, Matrix matrix)` : menghitung determinan berdasarkan ekspansi kofaktor pada baris x.
2. `ekspansiKofaktorKolom(int x, Matrix matrix)` : menghitung determinan berdasarkan ekspansi kofaktor pada kolom x.
3. `determinanReduksiBaris(Matrix matrix)`: menghitung determinan matriks persegi dengan metode reduksi baris. Matriks disalin, lalu diubah menjadi matriks segitiga atas dengan partial pivoting dan operasi pengurangan baris, sambil mencatat jumlah pertukaran baris. Mengembalikan null jika matriks tidak persegi dan 0 jika ada kolom tanpa pivot tidak nol.
4. `reduksiBaris(Matrix matrix, int countSwitch)`: menghitung determinan dari matriks segitiga atas dengan mengalikan semua elemen diagonalnya, lalu mengalikan hasilnya dengan $(-1)^{\text{countSwitch}}$ untuk memperhitungkan pertukaran baris.

d. Kelas Inverse

Kelas Inverse digunakan untuk menghitung matriks balikan dengan dua metode, yaitu augmentasi Gauss-Jordan dan metode adjoin.

Metode:

1. `inverseGaussJordan(Matrix matrix)` : menghitung invers dengan membentuk matriks augmented $[A|I]$, kemudian melakukan eliminasi Gauss-Jordan.
2. `inverseAdjoint(Matrix matrix)` : menghitung invers berdasarkan determinan dan matriks kofaktor yang ditransposisi menjadi matriks adjoin.
3. `findPivotPosition(...)` : mencari posisi pivot terbesar pada kolom yang sedang diproses.
4. `printAugmentedMatrix(...)` : menampilkan matriks augmented selama proses eliminasi.
5. `formatNumber(...)` : mengatur format tampilan angka.

e. Kelas `PolynomialInterpolation`

Kelas `PolynomialInterpolation` digunakan untuk melakukan perhitungan interpolasi polinomial berdasarkan titik-titik data yang diberikan.

Metode:

1. `interpolatePoly(Matrix data, double x)` : menghitung nilai polinomial pada suatu nilai x .
2. `printSteps(Matrix data, double x)` : menampilkan tahapan perhitungan interpolasi.
3. `getPolynomialCoefficients(Matrix data)` : memperoleh koefisien polinomial.
4. `printPolynomial(Matrix data)` : menampilkan persamaan polinomial hasil interpolasi.
5. `multiplyPolynomial(...)` : melakukan perkalian dua polinomial dalam representasi array koefisien.
6. `hasDuplicateX(Matrix data)` : memeriksa apakah terdapat nilai x yang sama.

f. Kelas `Handler`

Kelas `Handler` berfungsi sebagai penghubung antara pengguna dengan modul perhitungan. `Handler` menangani input, pemilihan metode, proses perhitungan, output, serta penyimpanan hasil ke file.

1. `SPLHandler`

- `run()` → menangani input SPL dari keyboard atau file, memilih metode penyelesaian, menjalankan perhitungan, dan menyimpan hasil.
- `readDataFromFile()` → membaca matriks augmented dari file.
- `writeOutputToFile()` → menyimpan hasil SPL ke file.

2. ModuleDeterminanHandler

- run() → menangani input matriks, memilih metode determinan, menampilkan langkah perhitungan, dan menyimpan hasil.
- readDataFromFile() → membaca matriks dari file.
- formatNumber() → mengatur format angka pada output.
- writeOutputToFile() → menyimpan hasil determinan ke file.

3. InverseHandler

- run() → menangani pemilihan metode invers, input matriks, proses perhitungan, dan output hasil.
- readMatrixFromFile() → membaca matriks dari file.
- writeOutputToFile() → menyimpan hasil matriks balikan ke file.

4. CubicInterpolationHandler

- run() → menangani input titik sampel, proses Natural Cubic Spline, evaluasi nilai x, dan penyimpanan hasil.
- readDataFromFile() → membaca titik sampel dari file.
- isXIncreasing() → memeriksa apakah nilai x terurut naik dan tidak memiliki nilai yang sama.
- formatNumber() → mengatur format angka.
- writeOutputToFile() → menyimpan hasil interpolasi ke file.

5. RegresiHandler

- run() → menangani input titik sampel dan knot, proses Regresi Spline Kubik, prediksi, dan penyimpanan hasil.
- readDataFromFile() → membaca data titik dari file.
- formatNumber() → mengatur format angka.
- writeOutputToFile() → menyimpan hasil regresi ke file.

g. Kelas Main

Kelas Main merupakan kelas utama yang menjadi titik awal eksekusi program. Kelas ini bertanggung jawab untuk menampilkan menu utama, menerima pilihan pengguna, serta mengarahkan proses ke modul sesuai fitur yang dipilih.

Atribut utama:

1. sc : objek Scanner yang digunakan untuk membaca input pengguna.
2. running : variabel boolean yang menentukan apakah program masih berjalan atau sudah dihentikan.
3. Metode utama:

4. `main()` : menjalankan keseluruhan program, menampilkan menu utama, membaca pilihan pengguna, melakukan validasi input, dan memanggil handler dari fitur yang dipilih.

Menu utama yang disediakan:

1. Sistem Persamaan Linier
2. Determinan
3. Matriks Balikan
4. Interpolasi Polinomial
5. Interpolasi Spline
6. Regresi Linier
7. Keluar

h. Kelas `CubicRegression`

Kelas `CubicRegression` digunakan untuk memodelkan hubungan antara variabel x dan y dengan regresi spline kubik.

Metode:

1. `regressX(Matrix titikSampel, int jumlahKnot, Matrix posisiKnot)` : Melakukan perhitungan regresi linear terhadap titik sampel berdasarkan knot.

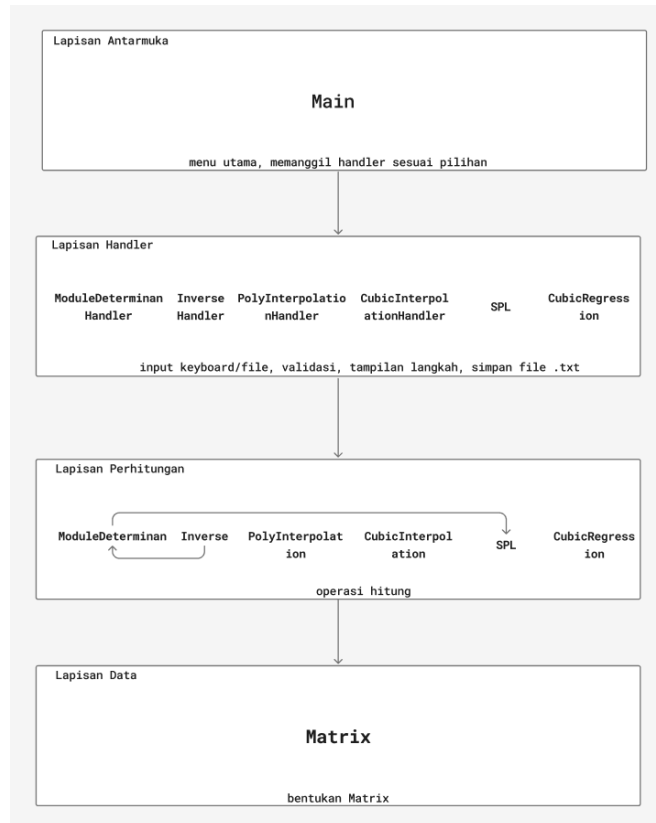
i. Kelas `CubicInterpolationFunction`

Kelas `ModuleDeterminan` digunakan untuk melakukan perhitungan persamaan segmen berdasarkan titik-titik diskrit yang dilalui

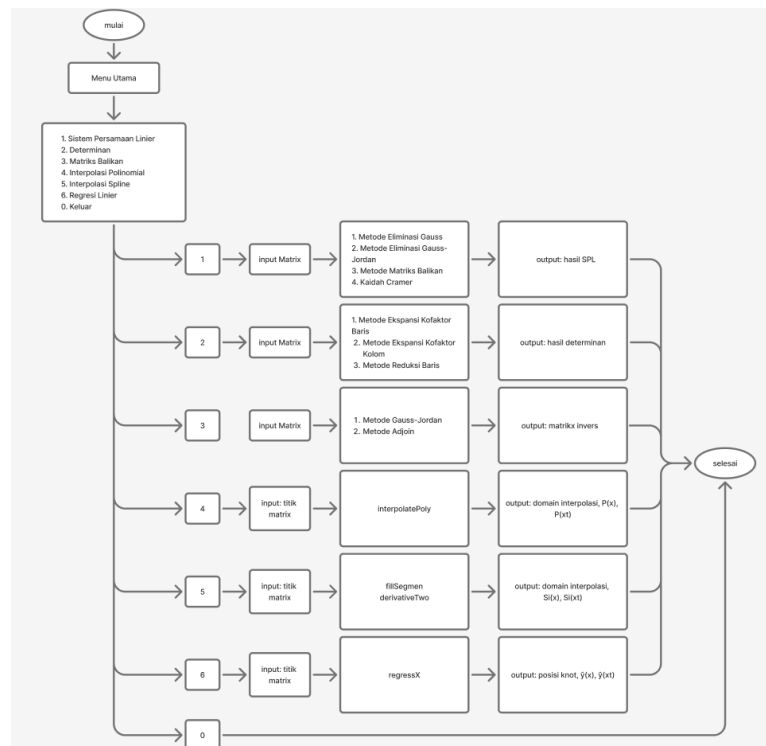
Metode:

1. `fillSegmen(Matrix matrix)` : menghitung koefisien a , b , c , d dan persamaan polinom kubik setiap segmen.
2. `derivativeTwo(Matrix S)` : menghitung koefisien turunan kedua setiap segmen.

B. Arsitektur Pustaka



C. Alur Program



D. Maven dan Pembuatan *.jar*

Apache Maven digunakan sebagai build automation tool dalam pengembangan program Java. Maven membantu mengelola proses pembangunan proyek, mulai dari kompilasi kode sumber hingga pengemasan program menjadi berkas JAR. Konfigurasi proyek Maven disimpan dalam file `pom.xml` yang memuat informasi dan konfigurasi yang diperlukan dalam proses build.

Setelah proses pengembangan selesai, program dapat dikompilasi dan dikemas menjadi JAR menggunakan perintah `mvn package`. Hasil proses tersebut akan disimpan pada folder `target/`. Berkas JAR kemudian dapat digunakan sebagai bentuk distribusi pustaka atau program Java yang telah dikompilasi.

IV. Eksperimen

A. Matriks Balikan (Invers)

Fitur	Matriks Balikan
Metode	Augmented Gauss-Jordan
Kasus Uji 1	$\begin{array}{cccc} 0 & 1 & 2 & -1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ -1 & 2 & 1 & 0 \end{array}$
Output	Matrix augmented $[A I]$: $\begin{array}{cccc cccc} 0 & 1 & 2 & -1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 2 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}$ $R1 \leftrightarrow R3$ $\begin{array}{cccc cccc} 2 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 & 0 & 0 \\ -1 & 2 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}$ $R1 \leftarrow (1/2)R1$ $\begin{array}{cccc cccc} 1 & 0.5 & 0 & 0.5 & 0 & 0 & 0.5 & 0 \end{array}$

	1	0	1	2	0	1	0	0
	0	1	2	-1	1	0	0	0
	-1	2	1	0	0	0	0	1
R2 = R2 - (1)R1								
	1	0.5	0	0.5	0	0	0.5	0
	0	-0.5	1	1.5	0	1	-0.5	0
	0	1	2	-1	1	0	0	0
	-1	2	1	0	0	0	0	1
R4 = R4 - (-1)R1								
	1	0.5	0	0.5	0	0	0.5	0
	0	-0.5	1	1.5	0	1	-0.5	0
	0	1	2	-1	1	0	0	0
	0	2.5	1	0.5	0	0	0.5	1
R2 <-> R4								
	1	0.5	0	0.5	0	0	0.5	0
	0	2.5	1	0.5	0	0	0.5	1
	0	1	2	-1	1	0	0	0
	0	-0.5	1	1.5	0	1	-0.5	0
R2 <- (1/2.5)R2								
	1	0.5	0	0.5	0	0	0.5	0
	0	1	0.4	0.2	0	0	0.2	0.4
	0	1	2	-1	1	0	0	0
	0	-0.5	1	1.5	0	1	-0.5	0
R1 = R1 - (0.5)R2								
	1	0	-0.2	0.4	0	0	0.4	-0.2
	0	1	0.4	0.2	0	0	0.2	0.4
	0	1	2	-1	1	0	0	0
	0	-0.5	1	1.5	0	1	-0.5	0
R3 = R3 - (1)R2								
	1	0	-0.2	0.4	0	0	0.4	-0.2
	0	1	0.4	0.2	0	0	0.2	0.4
	0	0	1.6	-1.2	1	0	-0.2	-0.4
	0	-0.5	1	1.5	0	1	-0.5	0
R4 = R4 - (-0.5)R2								
	1	0	-0.2	0.4	0	0	0.4	-0.2
	0	1	0.4	0.2	0	0	0.2	0.4
	0	0	1.6	-1.2	1	0	-0.2	-0.4
	0	0	1.2	1.6	0	1	-0.4	0.2
R3 <- (1/1.6)R3								
	1	0	-0.2	0.4	0	0	0.4	-0.2
	0	1	0.4	0.2	0	0	0.2	0.4

0.4 0.3 -0.2 -0.1

	-0.3 0.4 -0.1 0.2
Analisis Kasus Uji 1	Pada kasus uji 1, fitur Matriks Balikan menggunakan metode Augmented Gauss-Jordan dengan matriks berukuran 4×4 sebagai input. Matriks diproses menggunakan bentuk augmented $[A I]$ dan operasi baris elementer hingga diperoleh matriks identitas pada bagian kiri. Bagian kanan matriks augmented selanjutnya merupakan matriks balikan dari matriks input.
Kasus Uji 2	$\begin{matrix} 1 & 2 & 0 & -1 & 3 \\ 2 & -1 & 4 & 0 & 1 \\ 0 & 3 & 1 & 2 & -2 \\ 4 & 0 & 5 & 1 & 2 \\ 3 & 1 & 4 & -1 & 4 \end{matrix}$
Output	Matrix tidak memiliki balikan
Analisis Kasus Uji 2	Pada Uji 2, fitur Matriks Balikan menggunakan metode Augmented Gauss-Jordan dengan matriks input berukuran 5×5 . Matriks input kemudian diproses dalam bentuk augmented $[A I]$ menggunakan operasi baris elementer. Berdasarkan perhitungan, matriks input memiliki determinan 0 sehingga matriks tersebut merupakan matriks singular dan tidak memiliki matriks balikan.

Fitur	Matriks Balikan
Metode	Adjoint
Kasus Uji 1	$\begin{matrix} 0 & 1 & 2 & -1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ -1 & 2 & 1 & 0 \end{matrix}$
Output	=== Matriks Balikan === Metode: Adjoint Input Matriks: $\begin{matrix} 0 & 1 & 2 & -1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ -1 & 2 & 1 & 0 \end{matrix}$ Determinan = 20.0

	Matriks Kofaktor: $\begin{bmatrix} 4 & -2 & 8 & -6 \\ -2 & 4 & 6 & 8 \\ 8 & 6 & -4 & -2 \\ -6 & 8 & -2 & 4 \end{bmatrix}$ Matriks Adjoint: $\begin{bmatrix} 4 & -2 & 8 & -6 \\ -2 & 4 & 6 & 8 \\ 8 & 6 & -4 & -2 \\ -6 & 8 & -2 & 4 \end{bmatrix}$ Perhitungan Matriks Inverse: $A^{-1} = (1 / 20) * \text{Adjoint}$ $A^{-1} = (1 / 20) *$ $\begin{bmatrix} 4 & -2 & 8 & -6 \\ -2 & 4 & 6 & 8 \\ 8 & 6 & -4 & -2 \\ -6 & 8 & -2 & 4 \end{bmatrix}$ Pembagian setiap elemen dengan determinan: $\begin{bmatrix} 4/20 & -2/20 & 8/20 & -6/20 \\ -2/20 & -4/20 & 6/20 & 8/20 \\ 8/20 & 6/20 & -4/20 & -2/20 \\ -6/20 & 8/20 & -2/20 & 4/20 \end{bmatrix}$ Matriks Inverse: $\begin{bmatrix} 0.2 & -0.1 & 0.4 & -0.3 \\ -0.1 & -0.2 & 0.3 & 0.4 \\ 0.4 & 0.3 & -0.2 & -0.1 \\ -0.3 & 0.4 & -0.1 & 0.2 \end{bmatrix}$
Analisis Kasus Uji 1	Pada Kasus Uji 1, fitur Matriks Balikan diuji menggunakan metode Adjoint dengan matriks berukuran 4×4. Program menghitung determinan matriks terlebih dahulu sebagai syarat untuk menentukan apakah matriks memiliki balikan. Karena determinan tidak sama dengan nol, proses dilanjutkan dengan pembentukan matriks kofaktor, kemudian dilakukan transpose terhadap matriks kofaktor untuk memperoleh matriks adjoint. Selanjutnya, matriks invers diperoleh dengan membagi setiap elemen matriks adjoint dengan determinan matriks.
Kasus Uji 2	$\begin{bmatrix} 1 & 2 & 0 & -1 & 3 \\ 2 & -1 & 4 & 0 & 1 \\ 0 & 3 & 1 & 2 & -2 \\ 4 & 0 & 5 & 1 & 2 \\ 3 & 1 & 4 & -1 & 4 \end{bmatrix}$
Output	Matrix tidak memiliki balikan
Analisis Kasus Uji 2	Pada Kasus Uji 2, fitur Matriks Balikan diuji menggunakan metode Adjoint dengan matriks berukuran 5×5. Program terlebih dahulu menghitung determinan matriks. Hasil determinan bernilai nol, sehingga matriks merupakan matriks singular dan tidak memiliki balikan.

B. Interpolasi Polinomial

Fitur	Interpolasi Polinomial							
Metode	-							
Kasus Uji 1	x	0.2	0.4	0.6	0.8	1.0	1.2	1.4
	$f(x)$	0.182	0.336	0.470	0.588	0.693	0.788	0.875
	$x = 0.3$							
	$x = 0.55$							
	$x = 0.95$							
	$x = 1.38$							
Output	=== Interpolasi Polinom === Data titik: (0.2, 0.182) (0.4, 0.336) (0.6, 0.47) (0.8, 0.588) (1, 0.693) (1.2, 0.788) (1.4, 0.875) Nilai x = 0.3 Perhitungan interpolasi: metode : Interpolasi Polinomial dengan Eliminasi Gauss Derajat polinom = 6 Bentuk umum polinom: $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6$ Membentuk SPL dari titik data: $P(0.2) = 0.182$ $P(0.4) = 0.336$ $P(0.6) = 0.47$ $P(0.8) = 0.588$ $P(1.0) = 0.693$ $P(1.2) = 0.788$ $P(1.4) = 0.875$							

	Matriks augmented: $\begin{array}{cccccccc c} 1 & 0.2 & 0.04 & 0.008 & 0.002 & 0 & 0 & 0 & 0.182 \\ 1 & 0.4 & 0.16 & 0.064 & 0.026 & 0.01 & 0.004 & 0 & 0.336 \\ 1 & 0.6 & 0.36 & 0.216 & 0.13 & 0.078 & 0.047 & 0 & 0.47 \\ 1 & 0.8 & 0.64 & 0.512 & 0.41 & 0.328 & 0.262 & 0 & 0.588 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0.693 \\ 1 & 1.2 & 1.44 & 1.728 & 2.074 & 2.488 & 2.986 & 0 & 0.788 \\ 1 & 1.4 & 1.96 & 2.744 & 3.842 & 5.378 & 7.53 & 0 & 0.875 \end{array}$ Matriks Awal: $\begin{array}{cccccccccc} 1.0 & 0.2 & 0.04 & 0.008 & 0.002 & 0.0 & 0.0 & 0.0 & 0.182 \\ 1.0 & 0.4 & 0.16 & 0.064 & 0.026 & 0.01 & 0.004 & 0.0 & 0.336 \\ 1.0 & 0.6 & 0.36 & 0.216 & 0.13 & 0.078 & 0.047 & 0.0 & 0.47 \\ 1.0 & 0.8 & 0.64 & 0.512 & 0.41 & 0.328 & 0.262 & 0.0 & 0.588 \\ 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 0.693 \\ 1.0 & 1.2 & 1.44 & 1.728 & 2.074 & 2.488 & 2.986 & 0.0 & 0.788 \\ 1.0 & 1.4 & 1.96 & 2.744 & 3.842 & 5.378 & 7.53 & 0.0 & 0.875 \end{array}$ After subtracting row 2 with row 1 multiplied by 1.0: $\begin{array}{cccccccccc} 1.0 & 0.2 & 0.04 & 0.008 & 0.002 & 0.0 & 0.0 & 0.0 & 0.182 \\ 0.0 & 0.2 & 0.12 & 0.056 & 0.024 & 0.01 & 0.004 & 0.0 & 0.154 \\ 1.0 & 0.6 & 0.36 & 0.216 & 0.13 & 0.078 & 0.047 & 0.0 & 0.47 \\ 1.0 & 0.8 & 0.64 & 0.512 & 0.41 & 0.328 & 0.262 & 0.0 & 0.588 \\ 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 0.693 \\ 1.0 & 1.2 & 1.44 & 1.728 & 2.074 & 2.488 & 2.986 & 0.0 & 0.788 \\ 1.0 & 1.4 & 1.96 & 2.744 & 3.842 & 5.378 & 7.53 & 0.0 & 0.875 \end{array}$ After subtracting row 3 with row 1 multiplied by 1.0: $\begin{array}{cccccccccc} 1.0 & 0.2 & 0.04 & 0.008 & 0.002 & 0.0 & 0.0 & 0.0 & 0.182 \\ 0.0 & 0.2 & 0.12 & 0.056 & 0.024 & 0.01 & 0.004 & 0.0 & 0.154 \\ 0.0 & 0.4 & 0.32 & 0.208 & 0.128 & 0.077 & 0.047 & 0.0 & 0.288 \\ 1.0 & 0.8 & 0.64 & 0.512 & 0.41 & 0.328 & 0.262 & 0.0 & 0.588 \\ 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 0.693 \\ 1.0 & 1.2 & 1.44 & 1.728 & 2.074 & 2.488 & 2.986 & 0.0 & 0.788 \\ 1.0 & 1.4 & 1.96 & 2.744 & 3.842 & 5.378 & 7.53 & 0.0 & 0.875 \end{array}$ After subtracting row 4 with row 1 multiplied by 1.0: $\begin{array}{cccccccccc} 1.0 & 0.2 & 0.04 & 0.008 & 0.002 & 0.0 & 0.0 & 0.0 & 0.182 \\ 0.0 & 0.2 & 0.12 & 0.056 & 0.024 & 0.01 & 0.004 & 0.0 & 0.154 \\ 0.0 & 0.4 & 0.32 & 0.208 & 0.128 & 0.077 & 0.047 & 0.0 & 0.288 \\ 0.0 & 0.6 & 0.6 & 0.504 & 0.408 & 0.327 & 0.262 & 0.0 & 0.406 \\ 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 0.693 \\ 1.0 & 1.2 & 1.44 & 1.728 & 2.074 & 2.488 & 2.986 & 0.0 & 0.788 \\ 1.0 & 1.4 & 1.96 & 2.744 & 3.842 & 5.378 & 7.53 & 0.0 & 0.875 \end{array}$
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	After subtracting row 5 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 6 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 7 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After partial pivoting between row 2 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After normalizing row 2: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511
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	0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 3 with row 2 multiplied by 0.39999999999999997: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 4 with row 2 multiplied by 0.60000000000000001: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 5 with row 2 multiplied by 0.8: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 6 with row 2 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 7 with row 2 multiplied by 1.2: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
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	0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After partial pivoting between row 3 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After normalizing row 3: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 4 with row 3 multiplied by 0.24: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 5 with row 3 multiplied by 0.479999999999999987: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164
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	0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 6 with row 3 multiplied by 0.7999999999999998: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 7 with row 3 multiplied by 1.1999999999999997: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After partial pivoting between row 4 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After normalizing row 4: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After subtracting row 5 with row 4 multiplied by 0.19199999999999973: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
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	0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After subtracting row 6 with row 4 multiplied by 0.47999999999999965: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After subtracting row 7 with row 4 multiplied by 0.9599999999999991: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After partial pivoting between row 5 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After normalizing row 5: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004
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	0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After subtracting row 6 with row 5 multiplied by 0.1919999999999973: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After subtracting row 7 with row 5 multiplied by 0.5759999999999987: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001 0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004 After partial pivoting between row 6 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001 0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004 After normalizing row 6: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026 0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004 After subtracting row 7 with row 6 multiplied by 0.23040000000000282: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
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	0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026 0.0 0.0 0.0 0.0 0.0 0.0 0.046 -0.002 After normalizing row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026 0.0 0.0 0.0 0.0 0.0 0.0 1.0 -0.043 Koefisien: $a_0 = 0$ $a_1 = 1.003$ $a_2 = -0.546$ $a_3 = 0.479$ $a_4 = -0.408$ $a_5 = 0.208$ $a_6 = -0.043$ Persamaan interpolasi: $P(x) = -0.043x^6 + 0.208x^5 - 0.408x^4 + 0.479x^3 - 0.546x^2 + 1.003x$ Nilai interpolasi pada $x = 0.3$ $f(0.3) = 0.262$
	=== Interpolasi Polinom === Data titik: (0.2, 0.182) (0.4, 0.336) (0.6, 0.47) (0.8, 0.588)

(1, 0.693)
(1.2, 0.788)
(1.4, 0.875)

Nilai $x = 0.55$

Perhitungan interpolasi:
metode : Interpolasi Polinomial dengan Eliminasi
Gauss

Derajat polinom = 6

Bentuk umum polinom:

$$P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6$$

Membentuk SPL dari titik data:

$$P(0.2) = 0.182$$

$$P(0.4) = 0.336$$

$$P(0.6) = 0.47$$

$$P(0.8) = 0.588$$

$$P(1.0) = 0.693$$

$$P(1.2) = 0.788$$

$$P(1.4) = 0.875$$

Matriks augmented:

$$1 \ 0.2 \ 0.04 \ 0.008 \ 0.002 \ 0 \ 0 \ 0 \mid 0.182$$

$$1 \ 0.4 \ 0.16 \ 0.064 \ 0.026 \ 0.01 \ 0.004 \mid 0.336$$

$$1 \ 0.6 \ 0.36 \ 0.216 \ 0.13 \ 0.078 \ 0.047 \mid 0.47$$

$$1 \ 0.8 \ 0.64 \ 0.512 \ 0.41 \ 0.328 \ 0.262 \mid 0.588$$

$$1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \mid 0.693$$

$$1 \ 1.2 \ 1.44 \ 1.728 \ 2.074 \ 2.488 \ 2.986 \mid 0.788$$

$$1 \ 1.4 \ 1.96 \ 2.744 \ 3.842 \ 5.378 \ 7.53 \mid 0.875$$

Matriks Awal:

$$1.0 \ 0.2 \ 0.04 \ 0.008 \ 0.002 \ 0.0 \ 0.0 \ 0.182$$

$$1.0 \ 0.4 \ 0.16 \ 0.064 \ 0.026 \ 0.01 \ 0.004 \ 0.336$$

$$1.0 \ 0.6 \ 0.36 \ 0.216 \ 0.13 \ 0.078 \ 0.047 \ 0.47$$

$$1.0 \ 0.8 \ 0.64 \ 0.512 \ 0.41 \ 0.328 \ 0.262 \ 0.588$$

$$1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 0.693$$

$$1.0 \ 1.2 \ 1.44 \ 1.728 \ 2.074 \ 2.488 \ 2.986 \ 0.788$$

$$1.0 \ 1.4 \ 1.96 \ 2.744 \ 3.842 \ 5.378 \ 7.53 \ 0.875$$

After subtracting row 2 with row 1 multiplied by 1.0:

	1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 1.0 0.6 0.36 0.216 0.13 0.078 0.047 0.47 1.0 0.8 0.64 0.512 0.41 0.328 0.262 0.588 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 3 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 1.0 0.8 0.64 0.512 0.41 0.328 0.262 0.588 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 4 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 5 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 6 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875
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	After subtracting row 7 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After partial pivoting between row 2 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After normalizing row 2: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 3 with row 2 multiplied by 0.39999999999999997: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 4 with row 2 multiplied by 0.60000000000000001: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02
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	0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 5 with row 2 multiplied by 0.8: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 6 with row 2 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 7 with row 2 multiplied by 1.2: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After partial pivoting between row 3 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After normalizing row 3: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
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	0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 4 with row 3 multiplied by 0.24: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 5 with row 3 multiplied by 0.47999999999999987: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 6 with row 3 multiplied by 0.7999999999999998: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 7 with row 3 multiplied by 1.1999999999999997: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004
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	0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After partial pivoting between row 4 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After normalizing row 4: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After subtracting row 5 with row 4 multiplied by 0.1919999999999973: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After subtracting row 6 with row 4 multiplied by 0.4799999999999965: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069
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	After subtracting row 7 with row 4 multiplied by 0.9599999999999991: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After partial pivoting between row 5 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After normalizing row 5: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After subtracting row 6 with row 5 multiplied by 0.19199999999999973: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After subtracting row 7 with row 5 multiplied by 0.5759999999999987: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25
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	0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001 0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004
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After partial pivoting between row 6 and row 7:

	1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001 0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004
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After normalizing row 6:

	1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026 0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004
--	--

After subtracting row 7 with row 6 multiplied by
0.23040000000000282:

	1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026 0.0 0.0 0.0 0.0 0.0 0.0 0.046 -0.002
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After normalizing row 7:

	1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026 0.0 0.0 0.0 0.0 0.0 0.0 1.0 -0.043
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	Koefisien: $a_0 = 0$ $a_1 = 1.003$ $a_2 = -0.546$ $a_3 = 0.479$ $a_4 = -0.408$ $a_5 = 0.208$ $a_6 = -0.043$ Persamaan interpolasi: $P(x) = -0.043x^6 + 0.208x^5 - 0.408x^4 + 0.479x^3 - 0.546x^2 + 1.003x$ Nilai interpolasi pada $x = 0.55$ $f(0.55) = 0.438$
	=== Interpolasi Polinom === Data titik: (0.2, 0.182) (0.4, 0.336) (0.6, 0.47) (0.8, 0.588) (1, 0.693) (1.2, 0.788) (1.4, 0.875) Nilai $x = 0.95$ Perhitungan interpolasi: metode : Interpolasi Polinomial dengan Eliminasi Gauss Derajat polinom = 6 Bentuk umum polinom: $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6$ Membentuk SPL dari titik data:

$$P(0.2) = 0.182$$

$$P(0.4) = 0.336$$

$$P(0.6) = 0.47$$

$$P(0.8) = 0.588$$

$$P(1.0) = 0.693$$

$$P(1.2) = 0.788$$

$$P(1.4) = 0.875$$

Matriks augmented:

$$1 \ 0.2 \ 0.04 \ 0.008 \ 0.002 \ 0 \ 0 \ 0 \mid 0.182$$

$$1 \ 0.4 \ 0.16 \ 0.064 \ 0.026 \ 0.01 \ 0.004 \ 0 \mid 0.336$$

$$1 \ 0.6 \ 0.36 \ 0.216 \ 0.13 \ 0.078 \ 0.047 \ 0 \mid 0.47$$

$$1 \ 0.8 \ 0.64 \ 0.512 \ 0.41 \ 0.328 \ 0.262 \ 0 \mid 0.588$$

$$1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \mid 0.693$$

$$1 \ 1.2 \ 1.44 \ 1.728 \ 2.074 \ 2.488 \ 2.986 \ 0 \mid 0.788$$

$$1 \ 1.4 \ 1.96 \ 2.744 \ 3.842 \ 5.378 \ 7.53 \ 0 \mid 0.875$$

Matriks Awal:

$$1.0 \ 0.2 \ 0.04 \ 0.008 \ 0.002 \ 0.0 \ 0.0 \ 0.0 \mid 0.182$$

$$1.0 \ 0.4 \ 0.16 \ 0.064 \ 0.026 \ 0.01 \ 0.004 \ 0 \mid 0.336$$

$$1.0 \ 0.6 \ 0.36 \ 0.216 \ 0.13 \ 0.078 \ 0.047 \ 0 \mid 0.47$$

$$1.0 \ 0.8 \ 0.64 \ 0.512 \ 0.41 \ 0.328 \ 0.262 \ 0 \mid 0.588$$

$$1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 0 \mid 0.693$$

$$1.0 \ 1.2 \ 1.44 \ 1.728 \ 2.074 \ 2.488 \ 2.986 \ 0 \mid 0.788$$

$$1.0 \ 1.4 \ 1.96 \ 2.744 \ 3.842 \ 5.378 \ 7.53 \ 0 \mid 0.875$$

After subtracting row 2 with row 1 multiplied by 1.0:

$$1.0 \ 0.2 \ 0.04 \ 0.008 \ 0.002 \ 0.0 \ 0.0 \ 0 \mid 0.182$$

$$0.0 \ 0.2 \ 0.12 \ 0.056 \ 0.024 \ 0.01 \ 0.004 \ 0 \mid 0.154$$

$$1.0 \ 0.6 \ 0.36 \ 0.216 \ 0.13 \ 0.078 \ 0.047 \ 0 \mid 0.47$$

$$1.0 \ 0.8 \ 0.64 \ 0.512 \ 0.41 \ 0.328 \ 0.262 \ 0 \mid 0.588$$

$$1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 0 \mid 0.693$$

$$1.0 \ 1.2 \ 1.44 \ 1.728 \ 2.074 \ 2.488 \ 2.986 \ 0 \mid 0.788$$

$$1.0 \ 1.4 \ 1.96 \ 2.744 \ 3.842 \ 5.378 \ 7.53 \ 0 \mid 0.875$$

After subtracting row 3 with row 1 multiplied by 1.0:

$$1.0 \ 0.2 \ 0.04 \ 0.008 \ 0.002 \ 0.0 \ 0.0 \ 0 \mid 0.182$$

$$0.0 \ 0.2 \ 0.12 \ 0.056 \ 0.024 \ 0.01 \ 0.004 \ 0 \mid 0.154$$

$$0.0 \ 0.4 \ 0.32 \ 0.208 \ 0.128 \ 0.077 \ 0.047 \ 0 \mid 0.288$$

$$1.0 \ 0.8 \ 0.64 \ 0.512 \ 0.41 \ 0.328 \ 0.262 \ 0 \mid 0.588$$

$$1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0 \ 0 \mid 0.693$$

$$1.0 \ 1.2 \ 1.44 \ 1.728 \ 2.074 \ 2.488 \ 2.986 \ 0 \mid 0.788$$

$$1.0 \ 1.4 \ 1.96 \ 2.744 \ 3.842 \ 5.378 \ 7.53 \ 0 \mid 0.875$$

	After subtracting row 4 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 5 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 6 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 7 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After partial pivoting between row 2 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606
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	0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After normalizing row 2: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 3 with row 2 multiplied by 0.39999999999999997: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 4 with row 2 multiplied by 0.60000000000000001: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 5 with row 2 multiplied by 0.8: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 6 with row 2 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
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	0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 7 with row 2 multiplied by 1.2: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After partial pivoting between row 3 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After normalizing row 3: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 4 with row 3 multiplied by 0.24: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231
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	After subtracting row 5 with row 3 multiplied by 0.47999999999999987: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.048</td><td>0.096</td><td>0.125</td><td>0.134</td><td>0.004</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.192</td><td>0.422</td><td>0.614</td><td>0.753</td><td>0.015</td></tr><tr><td>0.0</td><td>0.0</td><td>0.8</td><td>1.44</td><td>1.952</td><td>2.438</td><td>2.966</td><td>-0.164</td></tr><tr><td>0.0</td><td>0.0</td><td>1.2</td><td>2.4</td><td>3.696</td><td>5.318</td><td>7.505</td><td>-0.231</td></tr></table> After subtracting row 6 with row 3 multiplied by 0.7999999999999998: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.048</td><td>0.096</td><td>0.125</td><td>0.134</td><td>0.004</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.192</td><td>0.422</td><td>0.614</td><td>0.753</td><td>0.015</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.48</td><td>1.152</td><td>1.862</td><td>2.58</td><td>0.036</td></tr><tr><td>0.0</td><td>0.0</td><td>1.2</td><td>2.4</td><td>3.696</td><td>5.318</td><td>7.505</td><td>-0.231</td></tr></table> After subtracting row 7 with row 3 multiplied by 1.1999999999999997: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.048</td><td>0.096</td><td>0.125</td><td>0.134</td><td>0.004</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.192</td><td>0.422</td><td>0.614</td><td>0.753</td><td>0.015</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.48</td><td>1.152</td><td>1.862</td><td>2.58</td><td>0.036</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.96</td><td>2.496</td><td>4.454</td><td>6.927</td><td>0.069</td></tr></table> After partial pivoting between row 4 and row 7: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.048</td><td>0.096</td><td>0.125</td><td>0.134</td><td>0.004</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.192</td><td>0.422</td><td>0.614</td><td>0.753</td><td>0.015</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.48</td><td>1.152</td><td>1.862</td><td>2.58</td><td>0.036</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.96</td><td>2.496</td><td>4.454</td><td>6.927</td><td>0.069</td></tr></table> After normalizing row 4: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr></table>	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	0.048	0.096	0.125	0.134	0.004	0.0	0.0	0.0	0.192	0.422	0.614	0.753	0.015	0.0	0.0	0.8	1.44	1.952	2.438	2.966	-0.164	0.0	0.0	1.2	2.4	3.696	5.318	7.505	-0.231	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	0.048	0.096	0.125	0.134	0.004	0.0	0.0	0.0	0.192	0.422	0.614	0.753	0.015	0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036	0.0	0.0	1.2	2.4	3.696	5.318	7.505	-0.231	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	0.048	0.096	0.125	0.134	0.004	0.0	0.0	0.0	0.192	0.422	0.614	0.753	0.015	0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036	0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	0.048	0.096	0.125	0.134	0.004	0.0	0.0	0.0	0.192	0.422	0.614	0.753	0.015	0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036	0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25
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After normalizing row 5:

1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182
0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77
0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25
0.0	0.0	0.0	1.0	2.0	2.6	2.8	0.083
0.0	0.0	0.0	0.0	1.0	3.0	5.6	-0.026
0.0	0.0	0.0	0.0	0.192	0.614	1.236	-0.004
0.0	0.0	0.0	0.0	0.576	1.958	4.239	-0.011

After subtracting row 6 with row 5 multiplied by
0.19199999999999973:

1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182
0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77
0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25
0.0	0.0	0.0	1.0	2.0	2.6	2.8	0.083
0.0	0.0	0.0	0.0	1.0	3.0	5.6	-0.026
0.0	0.0	0.0	0.0	0.0	0.038	0.161	0.001
0.0	0.0	0.0	0.0	0.576	1.958	4.239	-0.011

After subtracting row 7 with row 5 multiplied by
0.57599999999999987:

1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182
0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77
0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25
0.0	0.0	0.0	1.0	2.0	2.6	2.8	0.083
0.0	0.0	0.0	0.0	1.0	3.0	5.6	-0.026
0.0	0.0	0.0	0.0	0.0	0.038	0.161	0.001
0.0	0.0	0.0	0.0	0.0	0.23	1.014	0.004

After partial pivoting between row 6 and row 7:

1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182
0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77
0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25
0.0	0.0	0.0	1.0	2.0	2.6	2.8	0.083
0.0	0.0	0.0	0.0	1.0	3.0	5.6	-0.026
0.0	0.0	0.0	0.0	0.0	0.038	0.161	0.001
0.0	0.0	0.0	0.0	0.0	0.23	1.014	0.004

After normalizing row 6:

1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182
0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77
0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25

0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083
0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026
0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026
0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004

After subtracting row 7 with row 6 multiplied by
0.23040000000000282:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25
0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083
0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026
0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026
0.0 0.0 0.0 0.0 0.0 0.0 0.046 -0.002

After normalizing row 7:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25
0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083
0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026
0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026
0.0 0.0 0.0 0.0 0.0 0.0 1.0 -0.043

Koefisien:

$a_0 = 0$
 $a_1 = 1.003$
 $a_2 = -0.546$
 $a_3 = 0.479$
 $a_4 = -0.408$
 $a_5 = 0.208$
 $a_6 = -0.043$

Persamaan interpolasi:

$$P(x) = -0.043x^6 + 0.208x^5 - 0.408x^4 + 0.479x^3 - 0.546x^2 + 1.003x$$

Nilai interpolasi pada $x = 0.95$

$$f(0.95) = 0.668$$

=== Interpolasi Polinom ===

Data titik:

(0.2, 0.182)
(0.4, 0.336)
(0.6, 0.47)
(0.8, 0.588)
(1, 0.693)
(1.2, 0.788)
(1.4, 0.875)

Nilai $x = 1.38$

Perhitungan interpolasi:

metode : Interpolasi Polinomial dengan Eliminasi Gauss

Derajat polinom = 6

Bentuk umum polinom:

$$P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6$$

Membentuk SPL dari titik data:

$P(0.2) = 0.182$
 $P(0.4) = 0.336$
 $P(0.6) = 0.47$
 $P(0.8) = 0.588$
 $P(1.0) = 0.693$
 $P(1.2) = 0.788$
 $P(1.4) = 0.875$

Matriks augmented:

1 0.2 0.04 0.008 0.002 0 0 | 0.182
1 0.4 0.16 0.064 0.026 0.01 0.004 | 0.336
1 0.6 0.36 0.216 0.13 0.078 0.047 | 0.47
1 0.8 0.64 0.512 0.41 0.328 0.262 | 0.588
1 1 1 1 1 1 1 | 0.693
1 1.2 1.44 1.728 2.074 2.488 2.986 | 0.788
1 1.4 1.96 2.744 3.842 5.378 7.53 | 0.875

Matriks Awal:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
1.0 0.4 0.16 0.064 0.026 0.01 0.004 0.336

	1.0 0.6 0.36 0.216 0.13 0.078 0.047 0.47 1.0 0.8 0.64 0.512 0.41 0.328 0.262 0.588 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 2 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 1.0 0.6 0.36 0.216 0.13 0.078 0.047 0.47 1.0 0.8 0.64 0.512 0.41 0.328 0.262 0.588 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 3 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 1.0 0.8 0.64 0.512 0.41 0.328 0.262 0.588 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 4 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.693 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 5 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 1.0 1.2 1.44 1.728 2.074 2.488 2.986 0.788 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 6 with row 1 multiplied by 1.0:
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	1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 1.0 1.4 1.96 2.744 3.842 5.378 7.53 0.875 After subtracting row 7 with row 1 multiplied by 1.0: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After partial pivoting between row 2 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 0.2 0.12 0.056 0.024 0.01 0.004 0.154 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After normalizing row 2: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.4 0.32 0.208 0.128 0.077 0.047 0.288 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606 0.0 1.2 1.92 2.736 3.84 5.378 7.529 0.693 After subtracting row 3 with row 2 multiplied by 0.39999999999999997: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 0.08 0.096 0.08 0.058 0.039 -0.02 0.0 0.6 0.6 0.504 0.408 0.327 0.262 0.406 0.0 0.8 0.96 0.992 0.998 1.0 1.0 0.511 0.0 1.0 1.4 1.72 2.072 2.488 2.986 0.606
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	0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After normalizing row 3: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.24 0.336 0.336 0.298 0.25 -0.056 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 4 with row 3 multiplied by 0.24: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.48 0.768 0.902 0.96 0.984 -0.105 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 5 with row 3 multiplied by 0.47999999999999987: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.8 1.44 1.952 2.438 2.966 -0.164 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231 After subtracting row 6 with row 3 multiplied by 0.7999999999999998: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 0.048 0.096 0.125 0.134 0.004 0.0 0.0 0.0 0.192 0.422 0.614 0.753 0.015 0.0 0.0 0.0 0.48 1.152 1.862 2.58 0.036 0.0 0.0 1.2 2.4 3.696 5.318 7.505 -0.231
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	After subtracting row 7 with row 3 multiplied by 1.1999999999999997: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.048</td><td>0.096</td><td>0.125</td><td>0.134</td><td>0.004</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.192</td><td>0.422</td><td>0.614</td><td>0.753</td><td>0.015</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.48</td><td>1.152</td><td>1.862</td><td>2.58</td><td>0.036</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.96</td><td>2.496</td><td>4.454</td><td>6.927</td><td>0.069</td></tr></table> After partial pivoting between row 4 and row 7: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.048</td><td>0.096</td><td>0.125</td><td>0.134</td><td>0.004</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.192</td><td>0.422</td><td>0.614</td><td>0.753</td><td>0.015</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.48</td><td>1.152</td><td>1.862</td><td>2.58</td><td>0.036</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.96</td><td>2.496</td><td>4.454</td><td>6.927</td><td>0.069</td></tr></table> After normalizing row 4: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>1.0</td><td>2.0</td><td>2.6</td><td>2.8</td><td>0.083</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.192</td><td>0.422</td><td>0.614</td><td>0.753</td><td>0.015</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.48</td><td>1.152</td><td>1.862</td><td>2.58</td><td>0.036</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.96</td><td>2.496</td><td>4.454</td><td>6.927</td><td>0.069</td></tr></table> After subtracting row 5 with row 4 multiplied by 0.19199999999999973: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr><tr><td>0.0</td><td>0.0</td><td>1.0</td><td>1.2</td><td>1.0</td><td>0.72</td><td>0.482</td><td>-0.25</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>1.0</td><td>2.0</td><td>2.6</td><td>2.8</td><td>0.083</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td>0.038</td><td>0.115</td><td>0.215</td><td>-0.001</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.48</td><td>1.152</td><td>1.862</td><td>2.58</td><td>0.036</td></tr><tr><td>0.0</td><td>0.0</td><td>0.0</td><td>0.96</td><td>2.496</td><td>4.454</td><td>6.927</td><td>0.069</td></tr></table> After subtracting row 6 with row 4 multiplied by 0.47999999999999965: <table><tr><td>1.0</td><td>0.2</td><td>0.04</td><td>0.008</td><td>0.002</td><td>0.0</td><td>0.0</td><td>0.182</td></tr><tr><td>0.0</td><td>1.0</td><td>0.6</td><td>0.28</td><td>0.12</td><td>0.05</td><td>0.02</td><td>0.77</td></tr></table>	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	0.048	0.096	0.125	0.134	0.004	0.0	0.0	0.0	0.192	0.422	0.614	0.753	0.015	0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036	0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	0.048	0.096	0.125	0.134	0.004	0.0	0.0	0.0	0.192	0.422	0.614	0.753	0.015	0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036	0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	1.0	2.0	2.6	2.8	0.083	0.0	0.0	0.0	0.192	0.422	0.614	0.753	0.015	0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036	0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77	0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25	0.0	0.0	0.0	1.0	2.0	2.6	2.8	0.083	0.0	0.0	0.0	0.0	0.038	0.115	0.215	-0.001	0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036	0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069	1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182	0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77
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0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036																																																																																																																																																																																																																																										
0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069																																																																																																																																																																																																																																										
1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182																																																																																																																																																																																																																																										
0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77																																																																																																																																																																																																																																										
0.0	0.0	1.0	1.2	1.0	0.72	0.482	-0.25																																																																																																																																																																																																																																										
0.0	0.0	0.0	1.0	2.0	2.6	2.8	0.083																																																																																																																																																																																																																																										
0.0	0.0	0.0	0.0	0.038	0.115	0.215	-0.001																																																																																																																																																																																																																																										
0.0	0.0	0.0	0.48	1.152	1.862	2.58	0.036																																																																																																																																																																																																																																										
0.0	0.0	0.0	0.96	2.496	4.454	6.927	0.069																																																																																																																																																																																																																																										
1.0	0.2	0.04	0.008	0.002	0.0	0.0	0.182																																																																																																																																																																																																																																										
0.0	1.0	0.6	0.28	0.12	0.05	0.02	0.77																																																																																																																																																																																																																																										

	0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.96 2.496 4.454 6.927 0.069 After subtracting row 7 with row 4 multiplied by 0.9599999999999991: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After partial pivoting between row 5 and row 7: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 0.038 0.115 0.215 -0.001 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After normalizing row 5: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.192 0.614 1.236 -0.004 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011 After subtracting row 6 with row 5 multiplied by 0.19199999999999973: 1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182 0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77 0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25 0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001 0.0 0.0 0.0 0.0 0.576 1.958 4.239 -0.011
--	---

After subtracting row 7 with row 5 multiplied by
0.5759999999999987:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25
0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083
0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026
0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001
0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004

After partial pivoting between row 6 and row 7:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25
0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083
0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026
0.0 0.0 0.0 0.0 0.0 0.038 0.161 0.001
0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004

After normalizing row 6:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25
0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083
0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026
0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026
0.0 0.0 0.0 0.0 0.0 0.23 1.014 0.004

After subtracting row 7 with row 6 multiplied by
0.23040000000000282:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25
0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083
0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026
0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026
0.0 0.0 0.0 0.0 0.0 0.0 0.046 -0.002

After normalizing row 7:

1.0 0.2 0.04 0.008 0.002 0.0 0.0 0.182
0.0 1.0 0.6 0.28 0.12 0.05 0.02 0.77
0.0 0.0 1.0 1.2 1.0 0.72 0.482 -0.25

	0.0 0.0 0.0 1.0 2.0 2.6 2.8 0.083 0.0 0.0 0.0 0.0 1.0 3.0 5.6 -0.026 0.0 0.0 0.0 0.0 0.0 1.0 4.2 0.026 0.0 0.0 0.0 0.0 0.0 0.0 1.0 -0.043 Koefisien: a0 = 0 a1 = 1.003 a2 = -0.546 a3 = 0.479 a4 = -0.408 a5 = 0.208 a6 = -0.043 Persamaan interpolasi: P(x) = -0.043x^6 + 0.208x^5 - 0.408x^4 + 0.479x^3 - 0.546x^2 + 1.003x Nilai interpolasi pada x = 1.38 f(1.38) = 0.867														
Analisis Kasus Uji 1	Berdasarkan tujuh titik tersebut, program membentuk polinom interpolasi berderajat enam. Proses pembentukan polinom dilakukan dengan menyusun Sistem Persamaan Linear (SPL) dari substitusi setiap titik ke bentuk umum polinom, kemudian SPL diselesaikan menggunakan metode Eliminasi Gauss dengan partial pivoting. Hasil proses eliminasi menghasilkan koefisien polinom a_0 hingga a_6. Koefisien yang diperoleh kemudian digunakan untuk membentuk persamaan polinom interpolasi.														
Kasus Uji 2	<table><tr><td>t (menit)</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>T</td><td>25.0</td><td>31.2</td><td>38.6</td><td>47.5</td><td>58.1</td><td>70.8</td></tr></table> t = 1 T = ? t = 3.5 T = ? t = 7 T = ? t = 9.2 T = ?	t (menit)	0	2	4	6	8	10	T	25.0	31.2	38.6	47.5	58.1	70.8
t (menit)	0	2	4	6	8	10									
T	25.0	31.2	38.6	47.5	58.1	70.8									

	t = ? T = 65
Output	=== Interpolasi Polinom === Data titik: (0, 25) (2, 31.2) (4, 38.6) (6, 47.5) (8, 58.1) (10, 70.8) Nilai x = 1 Perhitungan interpolasi: metode : Interpolasi Polinomial dengan Eliminasi Gauss Derajat polinom = 5 Bentuk umum polinom: $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$ Membentuk SPL dari titik data: $P(0.0) = 25.0$ $P(2.0) = 31.2$ $P(4.0) = 38.6$ $P(6.0) = 47.5$ $P(8.0) = 58.1$ $P(10.0) = 70.8$ Matriks augmented: 1 0 0 0 0 0 25 1 2 4 8 16 32 31.2 1 4 16 64 256 1024 38.6 1 6 36 216 1296 7776 47.5 1 8 64 512 4096 32768 58.1 1 10 100 1000 10000 100000 70.8 Matriks Awal: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 1.0 2.0 4.0 8.0 16.0 32.0 31.2 1.0 4.0 16.0 64.0 256.0 1024.0 38.6 1.0 6.0 36.0 216.0 1296.0 7776.0 47.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8 After subtracting row 2 with row 1 multiplied by 1.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 1.0 4.0 16.0 64.0 256.0 1024.0 38.6

	1.0 6.0 36.0 216.0 1296.0 7776.0 47.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 3 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 1.0 6.0 36.0 216.0 1296.0 7776.0 47.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 4 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 5 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5 0.0 8.0 64.0 512.0 4096.0 32768.0 33.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 6 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5 0.0 8.0 64.0 512.0 4096.0 32768.0 33.1 0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8
--	--

After partial pivoting between row 2 and row 6:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5 0.0 8.0 64.0 512.0 4096.0 32768.0 33.1 0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8
--	--

After normalizing row 2:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
--	--

0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 3 with row 2 multiplied by 4.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 4 with row 2 multiplied by 6.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 5 with row 2 multiplied by 8.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 6 with row 2 multiplied by 10.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3
0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After partial pivoting between row 3 and row 6:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3
0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After normalizing row 3:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

	0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After subtracting row 4 with row 3 multiplied by 24.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 48.0 576.0 4800.0 0.3 0.0 0.0 48.0 480.0 4032.0 32640.0 8.3 0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After subtracting row 5 with row 3 multiplied by 48.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 48.0 576.0 4800.0 0.3 0.0 0.0 0.0 192.0 2688.0 26880.0 1.1 0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After subtracting row 6 with row 3 multiplied by 80.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 48.0 576.0 4800.0 0.3 0.0 0.0 0.0 192.0 2688.0 26880.0 1.1 0.0 0.0 0.0 480.0 7680.0 90240.0 2.8 After partial pivoting between row 4 and row 6: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 48.0 576.0 4800.0 0.3 0.0 0.0 0.0 192.0 2688.0 26880.0 1.1 0.0 0.0 0.0 480.0 7680.0 90240.0 2.8 After normalizing row 4: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 1.0 12.0 100.0 0.006 0.0 0.0 0.0 192.0 2688.0 26880.0 1.1 0.0 0.0 0.0 480.0 7680.0 90240.0 2.8 After subtracting row 5 with row 4 multiplied by 192.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 1.0 12.0 100.0 0.006 0.0 0.0 0.0 0.0 384.0 7680.0 -0.1 0.0 0.0 0.0 480.0 7680.0 90240.0 2.8
--	---

After subtracting row 6 with row 4 multiplied by 480.0:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After partial pivoting between row 5 and row 6:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After normalizing row 5:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 1.0 20.0 0.0
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After subtracting row 6 with row 5 multiplied by 1920.0:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 1.0 20.0 0.0
0.0 0.0 0.0 0.0 0.0 3840.0 0.3
```

After normalizing row 6:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 1.0 20.0 0.0
0.0 0.0 0.0 0.0 0.0 1.0 0.0
```

Koefisien:

```
a0 = 25
a1 = 2.892
a2 = 0.07
a3 = 0.02
a4 = -0.002
a5 = 0
```

	Persamaan interpolasi: $P(x) = 0x^5 - 0.002x^4 + 0.02x^3 + 0.07x^2 + 2.892x + 25$ Nilai interpolasi pada $x = 1$ $f(1) = 27.981$
	=== Interpolasi Polinom === Data titik: $(0, 25)$ $(2, 31.2)$ $(4, 38.6)$ $(6, 47.5)$ $(8, 58.1)$ $(10, 70.8)$ Nilai $x = 3.5$ Perhitungan interpolasi: metode : Interpolasi Polinomial dengan Eliminasi Gauss Derajat polinom = 5 Bentuk umum polinom: $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$ Membentuk SPL dari titik data: $P(0.0) = 25.0$ $P(2.0) = 31.2$ $P(4.0) = 38.6$ $P(6.0) = 47.5$ $P(8.0) = 58.1$ $P(10.0) = 70.8$ Matriks augmented: $1 \ 0 \ 0 \ 0 \ 0 \ 0 \ \ 25$ $1 \ 2 \ 4 \ 8 \ 16 \ 32 \ \ 31.2$ $1 \ 4 \ 16 \ 64 \ 256 \ 1024 \ \ 38.6$ $1 \ 6 \ 36 \ 216 \ 1296 \ 7776 \ \ 47.5$ $1 \ 8 \ 64 \ 512 \ 4096 \ 32768 \ \ 58.1$ $1 \ 10 \ 100 \ 1000 \ 10000 \ 100000 \ \ 70.8$ Matriks Awal: $1.0 \ 0.0 \ 0.0 \ 0.0 \ 0.0 \ 0.0 \ 25.0$ $1.0 \ 2.0 \ 4.0 \ 8.0 \ 16.0 \ 32.0 \ 31.2$ $1.0 \ 4.0 \ 16.0 \ 64.0 \ 256.0 \ 1024.0 \ 38.6$

	1.0 6.0 36.0 216.0 1296.0 7776.0 47.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 2 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 1.0 4.0 16.0 64.0 256.0 1024.0 38.6 1.0 6.0 36.0 216.0 1296.0 7776.0 47.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 3 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 1.0 6.0 36.0 216.0 1296.0 7776.0 47.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 4 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5 1.0 8.0 64.0 512.0 4096.0 32768.0 58.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 5 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5 0.0 8.0 64.0 512.0 4096.0 32768.0 33.1 1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8
--	--

After subtracting row 6 with row 1 multiplied by 1.0:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5 0.0 8.0 64.0 512.0 4096.0 32768.0 33.1 0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8
--	--

After partial pivoting between row 2 and row 6:

	1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 2.0 4.0 8.0 16.0 32.0 6.2 0.0 4.0 16.0 64.0 256.0 1024.0 13.6 0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
--	---

0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After normalizing row 2:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 3 with row 2 multiplied by 4.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 4 with row 2 multiplied by 6.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 5 with row 2 multiplied by 8.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 6 with row 2 multiplied by 10.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3
0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After partial pivoting between row 3 and row 6:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After normalizing row 3:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 24.0 192.0 1248.0 7680.0 3.9

0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After subtracting row 4 with row 3 multiplied by 24.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After subtracting row 5 with row 3 multiplied by 48.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 0.0 192.0 2688.0 26880.0 1.1

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After subtracting row 6 with row 3 multiplied by 80.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 0.0 192.0 2688.0 26880.0 1.1

0.0 0.0 0.0 480.0 7680.0 90240.0 2.8

After partial pivoting between row 4 and row 6:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 0.0 192.0 2688.0 26880.0 1.1

0.0 0.0 0.0 480.0 7680.0 90240.0 2.8

After normalizing row 4:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 1.0 12.0 100.0 0.006

0.0 0.0 0.0 192.0 2688.0 26880.0 1.1

0.0 0.0 0.0 480.0 7680.0 90240.0 2.8

After subtracting row 5 with row 4 multiplied by 192.0:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 480.0 7680.0 90240.0 2.8
```

After subtracting row 6 with row 4 multiplied by 480.0:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After partial pivoting between row 5 and row 6:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After normalizing row 5:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 1.0 20.0 0.0
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After subtracting row 6 with row 5 multiplied by 1920.0:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 1.0 20.0 0.0
0.0 0.0 0.0 0.0 0.0 3840.0 0.3
```

After normalizing row 6:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 1.0 20.0 0.0
0.0 0.0 0.0 0.0 0.0 1.0 0.0
```

	Koefisien: $\begin{aligned} a_0 &= 25 \\ a_1 &= 2.892 \\ a_2 &= 0.07 \\ a_3 &= 0.02 \\ a_4 &= -0.002 \\ a_5 &= 0 \end{aligned}$ Persamaan interpolasi: $P(x) = 0x^5 - 0.002x^4 + 0.02x^3 + 0.07x^2 + 2.892x + 25$ Nilai interpolasi pada $x = 3.5$ $f(3.5) = 36.617$
	=== Interpolasi Polinom === Data titik: $\begin{aligned} (0, 25) \\ (2, 31.2) \\ (4, 38.6) \\ (6, 47.5) \\ (8, 58.1) \\ (10, 70.8) \end{aligned}$ Nilai $x = 7$ Perhitungan interpolasi: metode : Interpolasi Polinomial dengan Eliminasi Gauss Derajat polinom = 5 Bentuk umum polinom: $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$ Membentuk SPL dari titik data: $\begin{aligned} P(0.0) &= 25.0 \\ P(2.0) &= 31.2 \\ P(4.0) &= 38.6 \\ P(6.0) &= 47.5 \\ P(8.0) &= 58.1 \\ P(10.0) &= 70.8 \end{aligned}$ Matriks augmented: $\begin{aligned} 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ \ 25 \\ 1 \ 2 \ 4 \ 8 \ 16 \ 32 \ \ 31.2 \end{aligned}$

1 4 16 64 256 1024 | 38.6
1 6 36 216 1296 7776 | 47.5
1 8 64 512 4096 32768 | 58.1
1 10 100 1000 10000 100000 | 70.8

Matriks Awal:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
1.0 2.0 4.0 8.0 16.0 32.0 31.2
1.0 4.0 16.0 64.0 256.0 1024.0 38.6
1.0 6.0 36.0 216.0 1296.0 7776.0 47.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 2 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
1.0 4.0 16.0 64.0 256.0 1024.0 38.6
1.0 6.0 36.0 216.0 1296.0 7776.0 47.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 3 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
1.0 6.0 36.0 216.0 1296.0 7776.0 47.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 4 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 5 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 6 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5

0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After partial pivoting between row 2 and row 6:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After normalizing row 2:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 3 with row 2 multiplied by 4.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 4 with row 2 multiplied by 6.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 5 with row 2 multiplied by 8.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 6 with row 2 multiplied by 10.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After partial pivoting between row 3 and row 6:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 8.0 48.0 224.0 960.0 1.2

0.0 0.0 24.0 192.0 1248.0 7680.0 3.9

0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After normalizing row 3:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 24.0 192.0 1248.0 7680.0 3.9

0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After subtracting row 4 with row 3 multiplied by 24.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After subtracting row 5 with row 3 multiplied by 48.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 0.0 192.0 2688.0 26880.0 1.1

0.0 0.0 80.0 960.0 9920.0 99840.0 14.8

After subtracting row 6 with row 3 multiplied by 80.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 0.0 192.0 2688.0 26880.0 1.1

0.0 0.0 0.0 480.0 7680.0 90240.0 2.8

After partial pivoting between row 4 and row 6:

1.0 0.0 0.0 0.0 0.0 0.0 25.0

0.0 1.0 2.0 4.0 8.0 16.0 3.1

0.0 0.0 1.0 6.0 28.0 120.0 0.15

0.0 0.0 0.0 48.0 576.0 4800.0 0.3

0.0 0.0 0.0 192.0 2688.0 26880.0 1.1

0.0 0.0 0.0 480.0 7680.0 90240.0 2.8

After normalizing row 4:

1.0	0.0	0.0	0.0	0.0	0.0	25.0
0.0	1.0	2.0	4.0	8.0	16.0	3.1
0.0	0.0	1.0	6.0	28.0	120.0	0.15
0.0	0.0	0.0	1.0	12.0	100.0	0.006
0.0	0.0	0.0	192.0	2688.0	26880.0	1.1
0.0	0.0	0.0	480.0	7680.0	90240.0	2.8

After subtracting row 5 with row 4 multiplied by 192.0:

1.0	0.0	0.0	0.0	0.0	0.0	25.0
0.0	1.0	2.0	4.0	8.0	16.0	3.1
0.0	0.0	1.0	6.0	28.0	120.0	0.15
0.0	0.0	0.0	1.0	12.0	100.0	0.006
0.0	0.0	0.0	0.0	384.0	7680.0	-0.1
0.0	0.0	0.0	480.0	7680.0	90240.0	2.8

After subtracting row 6 with row 4 multiplied by 480.0:

1.0	0.0	0.0	0.0	0.0	0.0	25.0
0.0	1.0	2.0	4.0	8.0	16.0	3.1
0.0	0.0	1.0	6.0	28.0	120.0	0.15
0.0	0.0	0.0	1.0	12.0	100.0	0.006
0.0	0.0	0.0	0.0	384.0	7680.0	-0.1
0.0	0.0	0.0	0.0	1920.0	42240.0	-0.2

After partial pivoting between row 5 and row 6:

1.0	0.0	0.0	0.0	0.0	0.0	25.0
0.0	1.0	2.0	4.0	8.0	16.0	3.1
0.0	0.0	1.0	6.0	28.0	120.0	0.15
0.0	0.0	0.0	1.0	12.0	100.0	0.006
0.0	0.0	0.0	0.0	384.0	7680.0	-0.1
0.0	0.0	0.0	0.0	1920.0	42240.0	-0.2

After normalizing row 5:

1.0	0.0	0.0	0.0	0.0	0.0	25.0
0.0	1.0	2.0	4.0	8.0	16.0	3.1
0.0	0.0	1.0	6.0	28.0	120.0	0.15
0.0	0.0	0.0	1.0	12.0	100.0	0.006
0.0	0.0	0.0	0.0	1.0	20.0	0.0
0.0	0.0	0.0	0.0	1920.0	42240.0	-0.2

After subtracting row 6 with row 5 multiplied by 1920.0:

1.0	0.0	0.0	0.0	0.0	0.0	25.0
0.0	1.0	2.0	4.0	8.0	16.0	3.1
0.0	0.0	1.0	6.0	28.0	120.0	0.15
0.0	0.0	0.0	1.0	12.0	100.0	0.006
0.0	0.0	0.0	0.0	1.0	20.0	0.0
0.0	0.0	0.0	0.0	0.0	3840.0	0.3

	After normalizing row 6: <pre> 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 1.0 12.0 100.0 0.006 0.0 0.0 0.0 0.0 1.0 20.0 0.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0 </pre> Koefisien: <pre> a0 = 25 a1 = 2.892 a2 = 0.07 a3 = 0.02 a4 = -0.002 a5 = 0 </pre> Persamaan interpolasi: $P(x) = 0x^5 - 0.002x^4 + 0.02x^3 + 0.07x^2 + 2.892x + 25$ Nilai interpolasi pada $x = 7$ $f(7) = 52.571$
	=== Interpolasi Polinom === Data titik: <pre> (0, 25) (2, 31.2) (4, 38.6) (6, 47.5) (8, 58.1) (10, 70.8) </pre> Nilai $x = 9.2$ Perhitungan interpolasi: metode : Interpolasi Polinomial dengan Eliminasi Gauss Derajat polinom = 5 Bentuk umum polinom: $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$ Membentuk SPL dari titik data: <pre> P(0.0) = 25.0 P(2.0) = 31.2 </pre>

$P(4.0) = 38.6$
 $P(6.0) = 47.5$
 $P(8.0) = 58.1$
 $P(10.0) = 70.8$

Matriks augmented:

1 0 0 0 0 0 | 25
1 2 4 8 16 32 | 31.2
1 4 16 64 256 1024 | 38.6
1 6 36 216 1296 7776 | 47.5
1 8 64 512 4096 32768 | 58.1
1 10 100 1000 10000 100000 | 70.8

Matriks Awal:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
1.0 2.0 4.0 8.0 16.0 32.0 31.2
1.0 4.0 16.0 64.0 256.0 1024.0 38.6
1.0 6.0 36.0 216.0 1296.0 7776.0 47.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 2 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
1.0 4.0 16.0 64.0 256.0 1024.0 38.6
1.0 6.0 36.0 216.0 1296.0 7776.0 47.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 3 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
1.0 6.0 36.0 216.0 1296.0 7776.0 47.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 4 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
1.0 8.0 64.0 512.0 4096.0 32768.0 58.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 5 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5

0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
1.0 10.0 100.0 1000.0 10000.0 100000.0 70.8

After subtracting row 6 with row 1 multiplied by 1.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After partial pivoting between row 2 and row 6:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 2.0 4.0 8.0 16.0 32.0 6.2
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After normalizing row 2:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 4.0 16.0 64.0 256.0 1024.0 13.6
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 3 with row 2 multiplied by 4.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 6.0 36.0 216.0 1296.0 7776.0 22.5
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 4 with row 2 multiplied by 6.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 8.0 64.0 512.0 4096.0 32768.0 33.1
0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8

After subtracting row 5 with row 2 multiplied by 8.0:

1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 8.0 48.0 224.0 960.0 1.2
0.0 0.0 24.0 192.0 1248.0 7680.0 3.9
0.0 0.0 48.0 480.0 4032.0 32640.0 8.3

	0.0 10.0 100.0 1000.0 10000.0 100000.0 45.8 After subtracting row 6 with row 2 multiplied by 10.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 8.0 48.0 224.0 960.0 1.2 0.0 0.0 24.0 192.0 1248.0 7680.0 3.9 0.0 0.0 48.0 480.0 4032.0 32640.0 8.3 0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After partial pivoting between row 3 and row 6: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 8.0 48.0 224.0 960.0 1.2 0.0 0.0 24.0 192.0 1248.0 7680.0 3.9 0.0 0.0 48.0 480.0 4032.0 32640.0 8.3 0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After normalizing row 3: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 24.0 192.0 1248.0 7680.0 3.9 0.0 0.0 48.0 480.0 4032.0 32640.0 8.3 0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After subtracting row 4 with row 3 multiplied by 24.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 48.0 576.0 4800.0 0.3 0.0 0.0 48.0 480.0 4032.0 32640.0 8.3 0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After subtracting row 5 with row 3 multiplied by 48.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 48.0 576.0 4800.0 0.3 0.0 0.0 0.0 192.0 2688.0 26880.0 1.1 0.0 0.0 80.0 960.0 9920.0 99840.0 14.8 After subtracting row 6 with row 3 multiplied by 80.0: 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 48.0 576.0 4800.0 0.3 0.0 0.0 0.0 192.0 2688.0 26880.0 1.1 0.0 0.0 0.0 480.0 7680.0 90240.0 2.8
--	--

After partial pivoting between row 4 and row 6:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 48.0 576.0 4800.0 0.3
0.0 0.0 0.0 192.0 2688.0 26880.0 1.1
0.0 0.0 0.0 480.0 7680.0 90240.0 2.8
```

After normalizing row 4:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 192.0 2688.0 26880.0 1.1
0.0 0.0 0.0 480.0 7680.0 90240.0 2.8
```

After subtracting row 5 with row 4 multiplied by 192.0:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 480.0 7680.0 90240.0 2.8
```

After subtracting row 6 with row 4 multiplied by 480.0:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After partial pivoting between row 5 and row 6:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 384.0 7680.0 -0.1
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

After normalizing row 5:

```
1.0 0.0 0.0 0.0 0.0 0.0 25.0
0.0 1.0 2.0 4.0 8.0 16.0 3.1
0.0 0.0 1.0 6.0 28.0 120.0 0.15
0.0 0.0 0.0 1.0 12.0 100.0 0.006
0.0 0.0 0.0 0.0 1.0 20.0 0.0
0.0 0.0 0.0 0.0 1920.0 42240.0 -0.2
```

	After subtracting row 6 with row 5 multiplied by 1920.0: <pre> 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 1.0 12.0 100.0 0.006 0.0 0.0 0.0 0.0 1.0 20.0 0.0 0.0 0.0 0.0 0.0 0.0 3840.0 0.3 </pre> After normalizing row 6: <pre> 1.0 0.0 0.0 0.0 0.0 0.0 25.0 0.0 1.0 2.0 4.0 8.0 16.0 3.1 0.0 0.0 1.0 6.0 28.0 120.0 0.15 0.0 0.0 0.0 1.0 12.0 100.0 0.006 0.0 0.0 0.0 0.0 1.0 20.0 0.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0 </pre> Koefisien: <pre> a0 = 25 a1 = 2.892 a2 = 0.07 a3 = 0.02 a4 = -0.002 a5 = 0 </pre> Persamaan interpolasi: $P(x) = 0x^5 - 0.002x^4 + 0.02x^3 + 0.07x^2 + 2.892x + 25$ Nilai interpolasi pada $x = 9.2$ $f(9.2) = 65.425$
	$T = 65$ maka $t = 8 < t < 10$
Analisis Kasus Uji 2	Pada studi kasus ini, program diuji menggunakan data pengukuran suhu komponen terhadap waktu, dengan enam titik pengukuran, yaitu $t = 0, 2, 4, 6, 8, 10$ menit dan suhu masing-masing sebesar $25.0, 31.2, 38.6, 47.5, 58.1, 70.8^{\circ}\text{C}$. Berdasarkan data tersebut, dibentuk polinom interpolasi yang melewati seluruh titik pengukuran. Program kemudian digunakan untuk memperkirakan suhu komponen pada waktu yang berada di antara titik pengukuran, yaitu $t = 1, t = 3.5, t = 7$, dan $t = 9.2$ menit.

C. Sistem Persamaan Linear

Fitur	Sistem Persamaan Linear
Metode	Eliminasi Gauss
Kasus Uji 1	$A = \begin{bmatrix} 0 & 2 & -1 & 1 \\ 1 & -1 & 2 & 3 \\ 2 & 1 & 1 & -1 \\ -1 & 3 & 0 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} -5 \\ 15 \\ 1 \\ -3 \end{bmatrix}$
Output	Pilih metode input matriks augmented: 1. Input manual 2. Input file Metode input: 1 ----- Sistem Persamaan Linear ----- Silahkan masukkan sistem persamaan linear anda dalam bentuk matriks augmented. Ukuran maksimal matriks augmented adalah 11*12, termasuk hasil baris. Apabila sudah selesai, silahkan tekan 'ENTER' pada baris kosong. Contoh input matriks augmented: <pre>1 2 3 4 5 6 7 8 9 10 11 12</pre> (Hasil input adalah matriks augmented berukuran 3*4) Input matriks augmented: <pre>0 2 -1 1 -5 1 -1 2 3 15 2 1 1 -1 1 -1 3 0 2 -3</pre> SPL dari matriks : <pre>2X3 + X4 = -5 1X2 + 2X3 + 3X4 = 15 2X2 + X3 - X4 = 1 -13X22X4 = -3</pre> Pilih metode penyelesaian:

1. Metode Eliminasi Gauss
2. Metode Eliminasi Gauss-Jordan
3. Metode Matriks Balikan
4. Kaidah Cramer

Metode penyelesaian yang dipilih:

1

Matriks Awal:

0.0 2.0 -1.0 1.0 -5.0
1.0 -1.0 2.0 3.0 15.0
2.0 1.0 1.0 -1.0 1.0
-1.0 3.0 0.0 2.0 -3.0

After partial pivoting between row 1 and row 3:

2.0 1.0 1.0 -1.0 1.0
1.0 -1.0 2.0 3.0 15.0
0.0 2.0 -1.0 1.0 -5.0
-1.0 3.0 0.0 2.0 -3.0

After normalizing row 1:

1.0 0.5 0.5 -0.5 0.5
1.0 -1.0 2.0 3.0 15.0
0.0 2.0 -1.0 1.0 -5.0
-1.0 3.0 0.0 2.0 -3.0

After subtracting row 2 with row 1 multiplied by 1.0:

1.0 0.5 0.5 -0.5 0.5
0.0 -1.5 1.5 3.5 14.5
0.0 2.0 -1.0 1.0 -5.0
-1.0 3.0 0.0 2.0 -3.0

After subtracting row 3 with row 1 multiplied by 0.0:

1.0 0.5 0.5 -0.5 0.5
0.0 -1.5 1.5 3.5 14.5
0.0 2.0 -1.0 1.0 -5.0
-1.0 3.0 0.0 2.0 -3.0

After subtracting row 4 with row 1 multiplied by -1.0:

1.0 0.5 0.5 -0.5 0.5
0.0 -1.5 1.5 3.5 14.5
0.0 2.0 -1.0 1.0 -5.0
0.0 3.5 0.5 1.5 -2.5

After partial pivoting between row 2 and row 4:

1.0 0.5 0.5 -0.5 0.5
0.0 3.5 0.5 1.5 -2.5
0.0 2.0 -1.0 1.0 -5.0
0.0 -1.5 1.5 3.5 14.5

After normalizing row 2:

1.0 0.5 0.5 -0.5 0.5
0.0 1.0 0.143 0.429 -0.714
0.0 2.0 -1.0 1.0 -5.0
0.0 -1.5 1.5 3.5 14.5

After subtracting row 3 with row 2 multiplied by 2.0:

1.0 0.5 0.5 -0.5 0.5
0.0 1.0 0.143 0.429 -0.714
0.0 0.0 -1.286 0.143 -3.571
0.0 -1.5 1.5 3.5 14.5

After subtracting row 4 with row 2 multiplied by -1.5:

1.0 0.5 0.5 -0.5 0.5
0.0 1.0 0.143 0.429 -0.714
0.0 0.0 -1.286 0.143 -3.571
0.0 0.0 1.714 4.143 13.429

After partial pivoting between row 3 and row 4:

1.0 0.5 0.5 -0.5 0.5
0.0 1.0 0.143 0.429 -0.714
0.0 0.0 1.714 4.143 13.429
0.0 0.0 -1.286 0.143 -3.571

After normalizing row 3:

1.0 0.5 0.5 -0.5 0.5
0.0 1.0 0.143 0.429 -0.714
0.0 0.0 1.0 2.417 7.833
0.0 0.0 -1.286 0.143 -3.571

After subtracting row 4 with row 3 multiplied by
-1.2857142857142856:

1.0 0.5 0.5 -0.5 0.5
0.0 1.0 0.143 0.429 -0.714
0.0 0.0 1.0 2.417 7.833
0.0 0.0 0.0 3.25 6.5

	After normalizing row 4: 1.0 0.5 0.5 -0.5 0.5 0.0 1.0 0.143 0.429 -0.714 0.0 0.0 1.0 2.417 7.833 0.0 0.0 0.0 1.0 2.0 SPL dari matriks : $10.5X_2 + 0.5X_3 - 0.5X_4 = 0.5$ $10.143X_3 + 0.429X_4 = -0.714$ $12.417X_4 = 7.833$ $1 = 2$ Sistem persamaan linear memiliki solusi: $X_1 = 1$; $X_2 = -2$; $X_3 = 3$; $X_4 = 2$ Nama file output (termasuk .txt): ujiKasus1.txt Hasil disimpan di ujiKasus1.txt
Analisis Kasus Uji 1	
Kasus Uji 2	
Output	
Analisis Kasus Uji 2	

D. Determinan

Fitur	Determinan
Metode	Ekspansi Kofaktor Baris
Kasus Uji 1	0 2 -1 3 4 1 2 -2 -2 3 0 1 1 -1 5 2
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah baris: 4

Masukkan jumlah kolom: 4

Masukkan matriks 4x4 (setiap baris berisi 4 elemen, dipisahkan spasi).

Pecahan boleh ditulis a/b, contoh: 1/2

0 2 -1 3

4 1 2 -2

-2 3 0 1

1 -1 5 2

Metode perhitungan determinan:

1. Ekspansi kofaktor baris

2. Ekspansi kofaktor kolom

3. Reduksi baris

Pilih: 1

Ekspansi sepanjang baris ke- (1 sampai 4): 1

Metode: Ekspansi kofaktor baris

Matriks input:

0 2 -1 3 4 1 2 -2 -2 3 0 1 1 -1 5 2

Langkah-langkah perhitungan:

Ekspansi kofaktor sepanjang baris ke-1

$\det(A) = \text{jumlah } a_{ij} * C_{ij}$, dengan $C_{ij} = (-1)^{(i+j)} * M_{ij}$

$a_{11} = 0$

Submatriks M11 (hapus baris 1 dan kolom 1):

1 2 -2 3 0 1 -1 5 2 $M_{11} = -49$

$C_{11} = (-1)^{(1+1)} * -49 = -49$

$a_{11} * C_{11} = 0 * -49 = 0$

$a_{12} = 2$

Submatriks M12 (hapus baris 1 dan kolom 2):

4 2 -2 3 0 1 1 5 2 $M_{12} = 10$

$C_{12} = (-1)^{(1+2)} * 10 = -10$

$a_{12} * C_{12} = 2 * -10 = -20$

$a_{13} = -1$

Submatriks M13 (hapus baris 1 dan kolom 3):

4 1 -2 3 0 1 1 -1 5 2 $M_{13} = 35$

$C_{13} = (-1)^{(1+3)} * 35 = 35$

$a_{13} * C_{13} = -1 * 35 = -35$

$a_{14} = 3$

Submatriks M14 (hapus baris 1 dan kolom 4):

4 1 2 -2 3 0 1 -1 5 2 $M_{14} = 68$

$C_{14} = (-1)^{(1+4)} * 68 = -68$

$a_{14} * C_{14} = 3 * -68 = -204$

	$\det(A) = (0) + (-20) + (-35) + (-204) = -259$ Hasil determinan: $\det(A) = -259$
Analisis Kasus Uji 1	Pada studi kasus ini, program diuji menggunakan ekspansikofaktor baris menghasilkan determinan -259
Kasus Uji 2	2 -1 3 0 4 1 2 -2 5 -1 3 1 1 2 0 0 4 -1 3 2 5 0 4 2 4
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah baris: 5 Masukkan jumlah kolom: 5 Masukkan matriks 5x5 (setiap baris berisi 5 elemen, dipisahkan spasi). Pecahan boleh ditulis a/b, contoh: 1/2 2 -1 3 0 4 1 2 -2 5 -1 3 1 1 2 0 0 4 -1 3 2 5 0 4 2 4 Metode perhitungan determinan: 1. Ekspansi kofaktor baris 2. Ekspansi kofaktor kolom 3. Reduksi baris Pilih: 1 Ekspansi sepanjang baris ke- (1 sampai 5): 1 Metode: Ekspansi kofaktor baris Matriks input: 2 -1 3 0 4 1 2 -2 5 -1 3 1 1 2 0 0 4 -1 3 2 5 0 4 2 4 Langkah-langkah perhitungan: Ekspansi kofaktor sepanjang baris ke-1 $\det(A) = \text{jumlah } a_{ij} * C_{ij}, \text{ dengan } C_{ij} = (-1)^{(i+j)} * M_{ij}$

	$a_{11} = 2$ Submatriks M11 (hapus baris 1 dan kolom 1): $2 \ 2 \ 5 \ -11 \ 1 \ 2 \ 04 \ -1 \ 3 \ 20 \ 4 \ 2 \ 4$ $M_{11} = -114$ $C_{11} = (-1)^{(1+1)} * -114 = -114$ $a_{11} * C_{11} = 2 * -114 = -228$ $a_{12} = -1$ Submatriks M12 (hapus baris 1 dan kolom 2): $1 \ -2 \ 5 \ -13 \ 1 \ 2 \ 00 \ -1 \ 3 \ 25 \ 4 \ 2 \ 4$ $M_{12} = -27$ $C_{12} = (-1)^{(1+2)} * -27 = 27$ $a_{12} * C_{12} = -1 * 27 = -27$ $a_{13} = 3$ Submatriks M13 (hapus baris 1 dan kolom 3): $1 \ 2 \ 5 \ -13 \ 1 \ 2 \ 00 \ 4 \ 3 \ 25 \ 0 \ 2 \ 4$ $M_{13} = 177$ $C_{13} = (-1)^{(1+3)} * 177 = 177$ $a_{13} * C_{13} = 3 * 177 = 531$ $a_{14} = 0$ Submatriks M14 (hapus baris 1 dan kolom 4): $1 \ 2 \ -2 \ -13 \ 1 \ 1 \ 00 \ 4 \ -1 \ 25 \ 0 \ 4 \ 4$ $M_{14} = -69$ $C_{14} = (-1)^{(1+4)} * -69 = 69$ $a_{14} * C_{14} = 0 * 69 = 0$ $a_{15} = 4$ Submatriks M15 (hapus baris 1 dan kolom 5): $1 \ 2 \ -2 \ 53 \ 1 \ 1 \ 20 \ 4 \ -1 \ 35 \ 0 \ 4 \ 2$ $M_{15} = -69$ $C_{15} = (-1)^{(1+5)} * -69 = -69$ $a_{15} * C_{15} = 4 * -69 = -276$ $\det(A) = (-228) + (-27) + (531) + (0) + (-276) = 0$ Hasil determinan: $\det(A) = 0$
Analisis Kasus Uji 2	Pada matrix kasus uji 2, determinan 0
Kasus Uji 3	$1/2 \ 1 \ 0 \ 1/2$ $1/2 \ 5/4 \ 1/4 \ 1/2$ $-1/10 \ -2/5 \ 1/5 \ -3/10$ $3/5 \ 5/4 \ 9/20 \ 3/5$
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah baris: 4

Masukkan jumlah kolom: 4

Masukkan matriks 4x4 (setiap baris berisi 4 elemen, dipisahkan spasi).

Pecahan boleh ditulis a/b, contoh: 1/2

1/2 1 0 1/2

1/2 5/4 1/4 1/2

-1/10 -2/5 1/5 -3/10

3/5 5/4 9/20 3/5

Metode perhitungan determinan:

1. Ekspansi kofaktor baris

2. Ekspansi kofaktor kolom

3. Reduksi baris

Pilih: 1

Ekspansi sepanjang baris ke- (1 sampai 4): 1

Metode: Ekspansi kofaktor baris

Matriks input:

0.5 1 0 0.5 0.5 1.25 0.25 0.5 -0.1 -0.4 0.2 -0.3 0.6 1.25 0.45
0.6

Langkah-langkah perhitungan:

Ekspansi kofaktor sepanjang baris ke-1

$\det(A) = \text{jumlah } a_{ij} * C_{ij}$, dengan $C_{ij} = (-1)^{(i+j)} * M_{ij}$

$a_{11} = 0.5$

Submatriks M11 (hapus baris 1 dan kolom 1):

1.25 0.25 0.5 -0.4 0.2 -0.3 1.25 0.45 0.6 $M_{11} = 0.07$

$C_{11} = (-1)^{(1+1)} * 0.07 = 0.07$

$a_{11} * C_{11} = 0.5 * 0.07 = 0.035$

$a_{12} = 1$

Submatriks M12 (hapus baris 1 dan kolom 2):

0.5 0.25 0.5 -0.1 0.2 -0.3 0.6 0.45 0.6 $M_{12} = 0.015$

$C_{12} = (-1)^{(1+2)} * 0.015 = -0.015$

$a_{12} * C_{12} = 1 * -0.015 = -0.015$

$a_{13} = 0$

Submatriks M13 (hapus baris 1 dan kolom 3):

0.5 1.25 0.5 -0.1 -0.4 -0.3 0.6 1.25 0.6 $M_{13} = -0.025$

$C_{13} = (-1)^{(1+3)} * -0.025 = -0.025$

$a_{13} * C_{13} = 0 * -0.025 = 0$

$a_{14} = 0.5$

Submatriks M14 (hapus baris 1 dan kolom 4):

0.5 1.25 0.25 -0.1 -0.4 0.2 0.6 1.25 0.45 $M_{14} = 0.02$

$C_{14} = (-1)^{(1+4)} * 0.02 = -0.02$

	$a_{14} * C_{14} = 0.5 * -0.02 = -0.01$ $\det(A) = (0.035) + (-0.015) + (0) + (-0.01) = 0.01$ Hasil determinan: $\det(A) = 0.01$
Analisis Kasus Uji 3	Pada kasus uji ini, menggunakan input pecahan, menghasilkan determinan 0.01

Fitur	Determinan
Metode	Reduksi Baris
Kasus Uji 1	0 2 -1 3 4 1 2 -2 -2 3 0 1 1 -1 5 2
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah baris: 4 Masukkan jumlah kolom: 4 Masukkan matriks 4x4 (setiap baris berisi 4 elemen, dipisahkan spasi). Pecahan boleh ditulis a/b, contoh: 1/2 0 2 -1 3 4 1 2 -2 -2 3 0 1 1 -1 5 2 Metode perhitungan determinan: 1. Ekspansi kofaktor baris 2. Ekspansi kofaktor kolom 3. Reduksi baris Pilih: 3 Metode: Reduksi baris Matriks input: 0 2 -1 3 4 1 2 -2 -2 3 0 1 1 -1 5 2

	Langkah-langkah perhitungan: Reduksi baris hingga matriks berbentuk segitiga atas. Pertukaran baris membalik tanda determinan, operasi $R_i - k \times R_j$ tidak mengubah determinan. Tukar baris 1 dengan baris 2: 4 1 2 -20 2 -1 3 -2 3 0 11 -1 5 2 $R_3 = R_3 - (-0.5) \times R_1$: 4 1 2 -20 2 -1 30 3.5 1 01 -1 5 2 $R_4 = R_4 - (0.25) \times R_1$: 4 1 2 -20 2 -1 30 3.5 1 00 -1.25 4.5 2.5 Tukar baris 2 dengan baris 3: 4 1 2 -20 3.5 1 00 2 -1 30 -1.25 4.5 2.5 $R_3 = R_3 - (0.5714) \times R_2$: 4 1 2 -20 3.5 1 00 0 -1.571 30 -1.25 4.5 2.5 $R_4 = R_4 - (-0.3571) \times R_2$: 4 1 2 -20 3.5 1 00 0 -1.571 30 0 4.857 2.5 Tukar baris 3 dengan baris 4: 4 1 2 -20 3.5 1 00 0 4.857 2.50 0 -1.571 3 $R_4 = R_4 - (-0.3235) \times R_3$: 4 1 2 -20 3.5 1 00 0 4.857 2.50 0 0 3.809 Jumlah pertukaran baris: 3 $\det(A) = (-1)^3 \times (4) \times (3.5) \times (4.8571) \times (3.8088) = -259$ Hasil determinan: $\det(A) = -259$
Analisis Kasus Uji 1	Pada studi kasus ini, program diuji menggunakan reduksi baris menghasilkan determinan -259
Kasus Uji 2	2 -1 3 0 4 1 2 -2 5 -1 3 1 1 2 0 0 4 -1 3 2 5 0 4 2 4
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah baris: 5 Masukkan jumlah kolom: 5 Masukkan matriks 5x5 (setiap baris berisi 5 elemen, dipisahkan spasi). Pecahan boleh ditulis a/b, contoh: 1/2

2 -1 3 0 4
 1 2 -2 5 -1
 3 1 1 2 0
 0 4 -1 3 2
 5 0 4 2 4

Metode perhitungan determinan:

1. Ekspansi kofaktor baris
2. Ekspansi kofaktor kolom
3. Reduksi baris

Pilih: 3

Metode: Reduksi baris

Matriks input:

2 -1 3 0 4 1 2 -2 5 -13 1 1 2 0 0 4 -1 3 25 0 4 2 4

Langkah-langkah perhitungan:

Reduksi baris hingga matriks berbentuk segitiga atas.

Pertukaran baris membalik tanda determinan, operasi $R_i - k \times R_j$ tidak mengubah determinan.

Tukar baris 1 dengan baris 5:

5 0 4 2 4 1 2 -2 5 -13 1 1 2 0 0 4 -1 3 22 -1 3 0 4

$R_2 = R_2 - (0.2) \times R_1$:

5 0 4 2 4 0 2 -2.8 4.6 -1.83 1 1 2 0 0 4 -1 3 22 -1 3 0 4

$R_3 = R_3 - (0.6) \times R_1$:

5 0 4 2 4 0 2 -2.8 4.6 -1.80 1 -1.4 0.8 -2.40 4 -1 3 22 -1 3 0 4

$R_5 = R_5 - (0.4) \times R_1$:

5 0 4 2 4 0 2 -2.8 4.6 -1.80 1 -1.4 0.8 -2.40 4 -1 3 20 -1 1.4 -0.8 2.4

Tukar baris 2 dengan baris 4:

5 0 4 2 4 0 4 -1 3 20 1 -1.4 0.8 -2.40 2 -2.8 4.6 -1.80 -1 1.4 -0.8 2.4

$R_3 = R_3 - (0.25) \times R_2$:

5 0 4 2 4 0 4 -1 3 20 0 -1.15 0.05 -2.90 2 -2.8 4.6 -1.80 -1 1.4 -0.8 2.4

$R_4 = R_4 - (0.5) \times R_2$:

5 0 4 2 4 0 4 -1 3 20 0 -1.15 0.05 -2.90 0 -2.3 3.1 -2.80 -1 1.4 -0.8 2.4

$R_5 = R_5 - (-0.25) \times R_2$:

5 0 4 2 4 0 4 -1 3 20 0 -1.15 0.05 -2.90 0 -2.3 3.1 -2.80 0 1.15 -0.05 2.9

Tukar baris 3 dengan baris 4:

5 0 4 2 4 0 4 -1 3 20 0 -2.3 3.1 -2.80 0 -1.15 0.05 -2.90 0 1.15 -0.05 2.9

$R_4 = R_4 - (0.5) \times R_3$:

5 0 4 2 4 0 4 -1 3 20 0 -2.3 3.1 -2.80 0 0 -1.5 -1.50 0 1.15 -0.05 2.9

	$R5 = R5 - (-0.5) \times R3:$ 5 0 4 2 40 4 -1 3 20 0 -2.3 3.1 -2.80 0 0 -1.5 -1.50 0 0 1.5 1.5 $R5 = R5 - (-1) \times R4:$ 5 0 4 2 40 4 -1 3 20 0 -2.3 3.1 -2.80 0 0 -1.5 -1.50 0 0 0 Kolom 5 tidak memiliki pivot tidak nol, sehingga $\det(A) = 0$. Hasil determinan: $\det(A) = 0$
Analisis Kasus Uji 2	Pada matrix kasus uji 2, determinan 0
Kasus Uji 3	1/2 1 0 1/2 1/2 5/4 1/4 1/2 -1/10 -2/5 1/5 -3/10 3/5 5/4 9/20 3/5
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah baris: 4 Masukkan jumlah kolom: 4 Masukkan matriks 4x4 (setiap baris berisi 4 elemen, dipisahkan spasi). Pecahan boleh ditulis a/b, contoh: 1/2 1/2 1 0 1/2 1/2 5/4 1/4 1/2 -1/10 -2/5 1/5 -3/10 3/5 5/4 9/20 3/5 Metode perhitungan determinan: 1. Ekspansi kofaktor baris 2. Ekspansi kofaktor kolom 3. Reduksi baris Pilih: 1 Ekspansi sepanjang baris ke- (1 sampai 4): 1 Metode: Ekspansi kofaktor baris Matriks input: 0.5 1 0 0.50.5 1.25 0.25 0.5-0.1 -0.4 0.2 -0.30.6 1.25 0.45 0.6 Langkah-langkah perhitungan: Ekspansi kofaktor sepanjang baris ke-1

	$\det(A) = \text{jumlah } a_{ij} * C_{ij}, \text{ dengan } C_{ij} = (-1)^{(i+j)} * M_{ij}$ $a_{11} = 0.5$ Submatriks M11 (hapus baris 1 dan kolom 1): $1.25 \ 0.25 \ 0.5-0.4 \ 0.2 \ -0.3 \ 1.25 \ 0.45 \ 0.6$ $M_{11} = 0.07$ $C_{11} = (-1)^{(1+1)} * 0.07 = 0.07$ $a_{11} * C_{11} = 0.5 * 0.07 = 0.035$ $a_{12} = 1$ Submatriks M12 (hapus baris 1 dan kolom 2): $0.5 \ 0.25 \ 0.5-0.1 \ 0.2 \ -0.3 \ 0.6 \ 0.45 \ 0.6$ $M_{12} = 0.015$ $C_{12} = (-1)^{(1+2)} * 0.015 = -0.015$ $a_{12} * C_{12} = 1 * -0.015 = -0.015$ $a_{13} = 0$ Submatriks M13 (hapus baris 1 dan kolom 3): $0.5 \ 1.25 \ 0.5-0.1 \ -0.4 \ -0.3 \ 0.6 \ 1.25 \ 0.6$ $M_{13} = -0.025$ $C_{13} = (-1)^{(1+3)} * -0.025 = -0.025$ $a_{13} * C_{13} = 0 * -0.025 = 0$ $a_{14} = 0.5$ Submatriks M14 (hapus baris 1 dan kolom 4): $0.5 \ 1.25 \ 0.25-0.1 \ -0.4 \ 0.2 \ 0.6 \ 1.25 \ 0.45$ $M_{14} = 0.02$ $C_{14} = (-1)^{(1+4)} * 0.02 = -0.02$ $a_{14} * C_{14} = 0.5 * -0.02 = -0.01$ $\det(A) = (0.035) + (-0.015) + (0) + (-0.01) = 0.01$ Hasil determinan: $\det(A) = 0.01$
Analisis Kasus Uji 3	Pada kasus uji ini, menggunakan input pecahan, menghasilkan determinan 0.01

Fitur	Determinan
Metode	Ekspansi Kofaktor Kolom
Kasus Uji 1	$0 \ 2 \ -1 \ 3$ $4 \ 1 \ 2 \ -2$ $-2 \ 3 \ 0 \ 1$ $1 \ -1 \ 5 \ 2$
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt

Pilih: 1

Masukkan jumlah baris: 4

Masukkan jumlah kolom: 4

Masukkan matriks 4x4 (setiap baris berisi 4 elemen, dipisahkan spasi).

Pecahan boleh ditulis a/b, contoh: 1/2

0 2 -1 3

4 1 2 -2

-2 3 0 1

1 -1 5 2

Metode perhitungan determinan:

1. Ekspansi kofaktor baris

2. Ekspansi kofaktor kolom

3. Reduksi baris

Pilih: 2

Ekspansi sepanjang kolom ke- (1 sampai 4): 1

Metode: Ekspansi kofaktor kolom

Matriks input:

0 2 -1 3 4 1 2 -2 -2 3 0 1 1 -1 5 2

Langkah-langkah perhitungan:

Ekspansi kofaktor sepanjang kolom ke-1

$\det(A) = \text{jumlah } a_{ij} * C_{ij}, \text{ dengan } C_{ij} = (-1)^{(i+j)} * M_{ij}$

$a_{11} = 0$

Submatriks M11 (hapus baris 1 dan kolom 1):

1 2 -2 3 0 1 -1 5 2 $M_{11} = -49$

$C_{11} = (-1)^{(1+1)} * -49 = -49$

$a_{11} * C_{11} = 0 * -49 = 0$

$a_{21} = 4$

Submatriks M21 (hapus baris 2 dan kolom 1):

2 -1 3 3 0 1 -1 5 2 $M_{21} = 42$

$C_{21} = (-1)^{(2+1)} * 42 = -42$

$a_{21} * C_{21} = 4 * -42 = -168$

$a_{31} = -2$

Submatriks M31 (hapus baris 3 dan kolom 1):

2 -1 3 1 2 -2 -1 5 2 $M_{31} = 49$

$C_{31} = (-1)^{(3+1)} * 49 = 49$

$a_{31} * C_{31} = -2 * 49 = -98$

$a_{41} = 1$

Submatriks M41 (hapus baris 4 dan kolom 1):

	$2 \cdot (-1)^{3+1} \cdot (-23) + (-1)^{4+1} \cdot (-1) = -7$ $C_{41} = (-1)^{4+1} \cdot (-7) = 7$ $a_{41} \cdot C_{41} = 1 \cdot 7 = 7$ $\det(A) = (0) + (-168) + (-98) + (7) = -259$ Hasil determinan: $\det(A) = -259$
Analisis Kasus Uji 1	Pada studi kasus ini, program diuji menggunakan reduksi kolom menghasilkan determinan -259
Kasus Uji 2	$\begin{matrix} 2 & -1 & 3 & 0 & 4 \\ 1 & 2 & -2 & 5 & -1 \\ 3 & 1 & 1 & 2 & 0 \\ 0 & 4 & -1 & 3 & 2 \\ 5 & 0 & 4 & 2 & 4 \end{matrix}$
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah baris: 5 Masukkan jumlah kolom: 5 Masukkan matriks 5x5 (setiap baris berisi 5 elemen, dipisahkan spasi). Pecahan boleh ditulis a/b, contoh: 1/2 $\begin{matrix} 2 & -1 & 3 & 0 & 4 \\ 1 & 2 & -2 & 5 & -1 \\ 3 & 1 & 1 & 2 & 0 \\ 0 & 4 & -1 & 3 & 2 \\ 5 & 0 & 4 & 2 & 4 \end{matrix}$ Metode perhitungan determinan: 1. Ekspansi kofaktor baris 2. Ekspansi kofaktor kolom 3. Reduksi baris Pilih: 2 Ekspansi sepanjang kolom ke- (1 sampai 5): 1 Metode: Ekspansi kofaktor kolom Matriks input: $\begin{matrix} 2 & -1 & 3 & 0 & 4 \\ 1 & 2 & -2 & 5 & -1 \\ 3 & 1 & 1 & 2 & 0 \\ 0 & 4 & -1 & 3 & 2 \\ 5 & 0 & 4 & 2 & 4 \end{matrix}$ Langkah-langkah perhitungan:

	Ekspansi kofaktor sepanjang kolom ke-1 $\det(A) = \text{jumlah } a_{ij} * C_{ij}$, dengan $C_{ij} = (-1)^{(i+j)} * M_{ij}$ $a_{11} = 2$ Submatriks M11 (hapus baris 1 dan kolom 1): $2 \ 2 \ 5 \ -11 \ 1 \ 2 \ 04 \ -1 \ 3 \ 20 \ 4 \ 2 \ 4$ $M_{11} = -114$ $C_{11} = (-1)^{(1+1)} * -114 = -114$ $a_{11} * C_{11} = 2 * -114 = -228$ $a_{21} = 1$ Submatriks M21 (hapus baris 2 dan kolom 1): $-1 \ 3 \ 0 \ 41 \ 1 \ 2 \ 04 \ -1 \ 3 \ 20 \ 4 \ 2 \ 4$ $M_{21} = 0$ $C_{21} = (-1)^{(2+1)} * 0 = 0$ $a_{21} * C_{21} = 1 * 0 = 0$ $a_{31} = 3$ Submatriks M31 (hapus baris 3 dan kolom 1): $-1 \ 3 \ 0 \ 42 \ -2 \ 5 \ -14 \ -1 \ 3 \ 20 \ 4 \ 2 \ 4$ $M_{31} = -114$ $C_{31} = (-1)^{(3+1)} * -114 = -114$ $a_{31} * C_{31} = 3 * -114 = -342$ $a_{41} = 0$ Submatriks M41 (hapus baris 4 dan kolom 1): $-1 \ 3 \ 0 \ 42 \ -2 \ 5 \ -11 \ 1 \ 2 \ 00 \ 4 \ 2 \ 4$ $M_{41} = 0$ $C_{41} = (-1)^{(4+1)} * 0 = 0$ $a_{41} * C_{41} = 0 * 0 = 0$ $a_{51} = 5$ Submatriks M51 (hapus baris 5 dan kolom 1): $-1 \ 3 \ 0 \ 42 \ -2 \ 5 \ -11 \ 1 \ 2 \ 04 \ -1 \ 3 \ 2$ $M_{51} = 114$ $C_{51} = (-1)^{(5+1)} * 114 = 114$ $a_{51} * C_{51} = 5 * 114 = 570$ $\det(A) = (-228) + (0) + (-342) + (0) + (570) = 0$ Hasil determinan: $\det(A) = 0$
Analisis Kasus Uji 2	Pada matrix kasus uji 2, determinan 0
Kasus Uji 3	$1/2 \ 1 \ 0 \ 1/2$ $1/2 \ 5/4 \ 1/4 \ 1/2$ $-1/10 \ -2/5 \ 1/5 \ -3/10$ $3/5 \ 5/4 \ 9/20 \ 3/5$
Output	=== Determinan Matriks === Sumber input: 1. Keyboard 2. File .txt

Pilih: 1

Masukkan jumlah baris: 4

Masukkan jumlah kolom: 4

Masukkan matriks 4x4 (setiap baris berisi 4 elemen, dipisahkan spasi).

Pecahan boleh ditulis a/b, contoh: 1/2

1/2 1 0 1/2

1/2 5/4 1/4 1/2

-1/10 -2/5 1/5 -3/10

3/5 5/4 9/20 3/5

Metode perhitungan determinan:

1. Ekspansi kofaktor baris

2. Ekspansi kofaktor kolom

3. Reduksi baris

Pilih: 2

Ekspansi sepanjang kolom ke- (1 sampai 4): 1

Metode: Ekspansi kofaktor kolom

Matriks input:

0.5 1 0 0.5 0.5 1.25 0.25 0.5 -0.1 -0.4 0.2 -0.3 0.6 1.25 0.45
0.6

Langkah-langkah perhitungan:

Ekspansi kofaktor sepanjang kolom ke-1

$\det(A) = \text{jumlah } a_{ij} * C_{ij}$, dengan $C_{ij} = (-1)^{(i+j)} * M_{ij}$

$a_{11} = 0.5$

Submatriks M11 (hapus baris 1 dan kolom 1):

1.25 0.25 0.5 -0.4 0.2 -0.3 1.25 0.45 0.6 $M_{11} = 0.07$

$C_{11} = (-1)^{(1+1)} * 0.07 = 0.07$

$a_{11} * C_{11} = 0.5 * 0.07 = 0.035$

$a_{21} = 0.5$

Submatriks M21 (hapus baris 2 dan kolom 1):

1 0 0.5 -0.4 0.2 -0.3 1.25 0.45 0.6 $M_{21} = 0.04$

$C_{21} = (-1)^{(2+1)} * 0.04 = -0.04$

$a_{21} * C_{21} = 0.5 * -0.04 = -0.02$

$a_{31} = -0.1$

Submatriks M31 (hapus baris 3 dan kolom 1):

1 0 0.5 1.25 0.25 0.5 1.25 0.45 0.6 $M_{31} = 0.05$

$C_{31} = (-1)^{(3+1)} * 0.05 = 0.05$

$a_{31} * C_{31} = -0.1 * 0.05 = -0.005$

$a_{41} = 0.6$

	Submatriks M41 (hapus baris 4 dan kolom 1): $1 \ 0 \ 0.51.25 \ 0.25 \ 0.5-0.4 \ 0.2 \ -0.3$ $M_{41} = 0$ $C_{41} = (-1)^{(4+1)} * 0 = 0$ $a_{41} * C_{41} = 0.6 * 0 = 0$ $\det(A) = (0.035) + (-0.02) + (-0.005) + (0) = 0.01$ Hasil determinan: $\det(A) = 0.01$
Analisis Kasus Uji 3	Pada kasus uji ini, menggunakan input pecahan, menghasilkan determinan 0.01

E. Cubic Interpolation

Fitur	Cubic Interpolation
Metode	Cubic Interpolation
Kasus Uji 1	1 2 2 3 3 4 4 5 5 6
Output	=== Interpolasi Splina Kubik Natural === Sumber input: 1. Keyboard 2. File .txt Pilih: 1 Masukkan jumlah titik: 5 Masukkan 5 pasang titik, satu titik per baris dengan format: x y Pecahan boleh ditulis a/b, contoh: 1/2 1 2 2 3 3 4 4 5 5 6 Matriks Awal: 1.0 0.0 0.0 0.0 0.0 0.0 1.0 4.0 1.0 0.0 0.0 0.0 0.0 1.0 4.0 1.0 0.0 0.0 0.0 0.0 1.0 4.0 1.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0

After subtracting row 2 with row 1 multiplied by 1.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 4.0 1.0 0.0 0.0 0.0
0.0 1.0 4.0 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 3 with row 1 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 4.0 1.0 0.0 0.0 0.0
0.0 1.0 4.0 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 4 with row 1 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 4.0 1.0 0.0 0.0 0.0
0.0 1.0 4.0 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 5 with row 1 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 4.0 1.0 0.0 0.0 0.0
0.0 1.0 4.0 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After normalizing row 2:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 1.0 4.0 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 3 with row 2 multiplied by 1.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 3.75 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 4 with row 2 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 3.75 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 5 with row 2 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 3.75 1.0 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After normalizing row 3:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 1.0 4.0 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 4 with row 3 multiplied by 1.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 3.733 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 5 with row 3 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 3.733 1.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After normalizing row 4:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 1.0 0.268 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 5 with row 4 multiplied by 0.0:

1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 1.0 0.268 0.0
0.0 0.0 0.0 0.0 1.0 0.0

After subtracting row 4 with row 5 multiplied by
0.26785714285714285:

1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0

After subtracting row 3 with row 5 multiplied by 0.0:

1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0

After subtracting row 2 with row 5 multiplied by 0.0:

1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0

After subtracting row 1 with row 5 multiplied by 0.0:

1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.267 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0

After subtracting row 3 with row 4 multiplied by
0.26666666666666666:

1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.0 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0

After subtracting row 2 with row 4 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.0 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 1 with row 4 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.25 0.0 0.0 0.0
0.0 0.0 1.0 0.0 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 2 with row 3 multiplied by 0.25:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.0 0.0 0.0 0.0
0.0 0.0 1.0 0.0 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 1 with row 3 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.0 0.0 0.0 0.0
0.0 0.0 1.0 0.0 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

After subtracting row 1 with row 2 multiplied by 0.0:

```
1.0 0.0 0.0 0.0 0.0 0.0
0.0 1.0 0.0 0.0 0.0 0.0
0.0 0.0 1.0 0.0 0.0 0.0
0.0 0.0 0.0 1.0 0.0 0.0
0.0 0.0 0.0 0.0 1.0 0.0
```

Data titik:

1 22 33 44 55 6

Domain interpolasi: [1, 5]

Persamaan setiap segmen:

Segmen ke-0, domain [1, 2]

	$S_0(x) = 2 + 1(x - 1) + 0(x - 1)^2 + 0(x - 1)^3$ atau $S_0(x) = 1 + 1x + 0x^2 + 0x^3$ Segmen ke-1, domain [2, 3] $S_1(x) = 3 + 1(x - 2) + 0(x - 2)^2 + 0(x - 2)^3$ atau $S_1(x) = 1 + 1x + 0x^2 + 0x^3$ Segmen ke-2, domain [3, 4] $S_2(x) = 4 + 1(x - 3) + 0(x - 3)^2 + 0(x - 3)^3$ atau $S_2(x) = 1 + 1x + 0x^2 + 0x^3$ Segmen ke-3, domain [4, 5] $S_3(x) = 5 + 1(x - 4) + 0(x - 4)^2 + 0(x - 4)^3$ atau $S_3(x) = 1 + 1x + 0x^2 + 0x^3$ Nilai turunan kedua pada setiap knot: $S''(x_i) = 2c_i$ $S''(x_0) = S''(1) = 0$ $S''(x_1) = S''(2) = 0$ $S''(x_2) = S''(3) = 0$ $S''(x_3) = S''(4) = 0$ $S''(x_4) = S''(5) = 0$ Masukkan nilai x yang ingin dievaluasi: 2.3 Hasil evaluasi: x = 2.3 berada pada segmen S1 $S(2.3) = S_1(2.3) = 3.3$
Analisis Kasus Uji 1	Pada kasus uji ini, 5 titik diskrit dihitung persamaan segmen, ditemukan hasil turunan kedua, dan nilai $S(x_t)$ dengan x_t memenuhi domain, $x_t = 2.3$

V. Penutup

Berdasarkan proses perancangan, implementasi, dan pengujian yang telah dilakukan, Tugas Besar 1 Aljabar Linier dan Geometri berhasil menghasilkan sebuah pustaka komputasi berbasis Java yang mengimplementasikan berbagai metode perhitungan aljabar linier. Fitur yang dikembangkan meliputi penyelesaian Sistem Persamaan Linier (SPL),

perhitungan determinan, matriks balikan, interpolasi polinomial, Natural Cubic Spline, dan regresi spline kubik.

Pengembangan selanjutnya dapat dilakukan dengan meningkatkan validasi terhadap berbagai kemungkinan input dan kasus khusus agar program menjadi lebih robust. Selain itu, pengujian dapat diperluas dengan menggunakan lebih banyak variasi kasus, termasuk kasus batas dan input yang tidak valid.

VI. Referensi

[1] Institut Teknologi Bandung, IF2123 Aljabar Linier dan Geometri: Tugas Besar 1 Semester I 2026/2027, Bandung, Indonesia: Institut Teknologi Bandung, 2026.

[2] G. Munthe, *Topics in Polynomial Interpolation Theory*, Ph.D. dissertation, Centre of Mathematics for Applications, University of Oslo, Oslo, Norway, Dec. 2010. [Online]. Available:
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[3] R. L. Burden and J. D. Faires, *Numerical Analysis*, 9th ed. Boston, MA, USA: Brooks/Cole, Cengage Learning, 2011, ch. 3, sec. 3.5. [Online]. Available:
https://faculty.ksu.edu.sa/sites/default/files/numerical_analysis_9th.pdf

VII. Lampiran

Repository: <https://github.com/Lapdum/algeo26-tb1-InderoopsieArnes>

Tugas ini disusun sepenuhnya tanpa bantuan kecerdasan buatan (Generative AI), melainkan hasil pemikiran dan analisis mandiri.



Cendra Asih Chairunnisa



Nadia Layla Safira

A handwritten signature in black ink, appearing to read 'Necia', with a horizontal line drawn underneath it. The signature is set against a light gray, textured background.

Necia Aurely Greva Dedevi